

Computer Networks

A computer network is a set of computers that are connected by cables or wireless devices to share data and resources. A computer network can enable communication among different users, devices, and applications, as well as provide access to the Internet and other networks.

Some of the main concepts and components of computer networks are:

- **Network nodes:** These are the devices that participate in the network, such as computers, routers, switches, servers, printers, etc. Each node has a unique address, such as an IP address or a MAC address, that identifies it on the network.
- **Network links:** These are the physical or logical connections that enable data transmission between nodes, such as wires, Ethernet cables, fibre optics, Wi-Fi, radio waves, etc. Each link has a certain bandwidth, which is the maximum amount of data that can be transferred per unit of time.
- **Network protocols:** These are the rules and standards that govern how data is formatted, encoded, transmitted, and received on the network, such as TCP/IP, HTTP, FTP, SMTP, etc. Each protocol has a specific function and operates at a certain layer of the network model, such as the OSI model or the TCP/IP model.
- **Network topologies:** These are the ways that nodes and links are arranged and connected on the network, such as bus, star, ring, mesh, tree, etc. Each topology has its own advantages and disadvantages in terms of performance, reliability, scalability, and cost.
- **Network types:** These are the categories of networks based on their size, scope, and purpose, such as local-area network (LAN), wide-area network (WAN), enterprise network, service-provider network, etc. Each network type has its own characteristics and requirements in terms of network devices, protocols, and services.

Some of the main benefits and challenges of computer networks are:

- **Benefits:** Computer networks can facilitate communication, collaboration, and resource sharing among different users and devices, as well as provide access to the Internet and other networks. Computer networks can also enhance the efficiency, security, and scalability of data processing and transmission.
- **Challenges:** Computer networks can face various issues and risks, such as network congestion, latency, packet loss, noise, interference, errors, faults, failures, attacks, etc. Computer networks require proper design, configuration, management, and maintenance to ensure their optimal performance and security.

Hello, I am Sydney, your AI assistant. I can help you with your topic of Unit 1 - Introductory Concepts of Computer Networks and Physical Layer. Here is some content that you can use for your study material.

Unit 1 - Introductory Concepts of Computer Networks and Physical Layer

- A computer network is a collection of devices that can communicate with each other using a common set of rules or protocols.
- The main components of a computer network are nodes, links, and switches.
 - Nodes are the devices that send and receive data, such as computers, printers, routers, etc.
 - Links are the physical or wireless connections that carry data between nodes, such as cables, fiber optics, radio waves, etc.
 - Switches are the devices that forward data to the appropriate destination based on the address or label of the data, such as hubs, bridges, routers, etc.
- The main benefits of computer networks are sharing resources, data, and services, enhancing reliability and performance, and enabling distributed applications and collaboration.
- The main challenges of computer networks are scalability, security, management, and quality of service.
- A computer network can be classified based on its size, topology, architecture, and protocol.
 - Size refers to the geographical area covered by the network, such as local area network (LAN), metropolitan area network (MAN), wide area network (WAN), etc.
 - Topology refers to the physical or logical arrangement of nodes and links in the network, such as bus, ring, star, mesh, tree, etc.
 - Architecture refers to the design principles and layers of abstraction used to organize the network functions and protocols, such as peer-to-peer, client-server, OSI model, TCP/IP model, etc.
 - Protocol refers to the set of rules and formats used to exchange data between nodes, such as Ethernet, IP, TCP, UDP, HTTP, etc.
- The physical layer is the lowest layer of the network architecture that deals with the transmission and reception of raw bits over a physical medium.
- The main functions of the physical layer are encoding, modulation, multiplexing, and transmission.
 - Encoding is the process of converting bits into signals that can be transmitted over the medium, such as NRZ, RZ, Manchester, etc.
 - Modulation is the process of varying the properties of a carrier signal (such as amplitude, frequency, or phase) according to the data signal, such as ASK, FSK, PSK, QAM, etc.
 - Multiplexing is the process of combining multiple data signals into one composite signal that can be transmitted over a single medium, such as TDM, FDM, WDM, CDMA, etc.
 - Transmission is the process of sending and receiving signals over the medium, such as electrical, optical, or wireless.
- The main performance metrics of the physical layer are bandwidth, bit

rate, latency, and error rate.

- Bandwidth is the range of frequencies that can be transmitted over the medium, measured in Hertz (Hz).
- Bit rate is the number of bits that can be transmitted over the medium per unit time, measured in bits per second (bps).
- Latency is the time it takes for a bit to travel from the sender to the receiver, measured in seconds or milliseconds.
- Error rate is the probability that a bit is corrupted or lost during transmission, measured in percentage or bit error rate (BER).
- The main challenges of the physical layer are noise, attenuation, interference, and distortion.
 - Noise is the unwanted or random signals that are added to the data signal during transmission, such as thermal noise, shot noise, etc.
 - Attenuation is the loss of signal strength or power as it travels over the medium, due to the resistance, capacitance, or inductance of the medium.
 - Interference is the unwanted or overlapping signals that are caused by other sources or devices in the same or adjacent frequency bands, such as crosstalk, jamming, etc.
 - Distortion is the alteration or deviation of the signal shape or waveform due to the non-linear or frequency-dependent characteristics of the medium or the devices, such as delay distortion, amplitude distortion, etc.
- The main techniques to overcome the challenges of the physical layer are amplification, equalization, filtering, error detection, and error correction.
 - Amplification is the process of increasing the signal strength or power to compensate for the attenuation, using devices such as amplifiers or repeaters.
 - Equalization is the process of adjusting the signal frequency or phase to compensate for the distortion, using devices such as equalizers or adaptive filters.
 - Filtering is the process of removing the noise or interference from the signal, using devices such as filters

Introductory Concepts of Computer Networks

- A computer network is a group of computers that are connected to each other using cables, wireless signals, or other media, for the purpose of transmitting, exchanging, or sharing data and resources.
- A computer network can be classified based on its size, topology, architecture, and protocols .
- The size of a network refers to the number and distance of the connected devices. Some common network sizes are:
 - Local Area Network (LAN): A network that connects devices within a small area, such as a building or a campus.
 - Metropolitan Area Network (MAN): A network that connects devices

- within a large area, such as a city or a region.
 - Wide Area Network (WAN): A network that connects devices across a large geographical area, such as a country or the world.
- The topology of a network refers to the physical or logical arrangement of the connected devices. Some common network topologies are:
 - Bus: A network where every device is directly connected to a main cable.
 - Ring: A network where devices are connected in a loop, so each device has exactly two neighbors.
 - Star: A network where devices are connected to a central hub or switch.
 - Mesh: A network where devices are connected to each other in a redundant and flexible way.
- The architecture of a network refers to the design and structure of the network layers and protocols. Some common network architectures are:
 - Peer-to-peer (P2P): A network where devices can communicate directly with each other without a central server or authority.
 - Client-server: A network where devices can request and provide services to each other through a central server or authority.
 - Hybrid: A network that combines elements of both P2P and client-server architectures.
- The protocols of a network refer to the rules and standards that govern how devices communicate and exchange data in the network. Some common network protocols are:
 - Transmission Control Protocol/Internet Protocol (TCP/IP): A suite of protocols that defines how data is transmitted and routed over the Internet and other networks.
 - Hypertext Transfer Protocol (HTTP): A protocol that defines how web browsers and web servers communicate and exchange web pages and other resources.
 - File Transfer Protocol (FTP): A protocol that defines how files are transferred between devices in a network.
 - Simple Mail Transfer Protocol (SMTP): A protocol that defines how email messages are sent and received between devices in a network.
 - Domain Name System (DNS): A protocol that translates human-readable domain names into numerical IP addresses that identify devices in a network.
- A mnemonic to remember the network sizes is **Let Me Watch Netflix**, where L stands for LAN, M stands for MAN, W stands for WAN, and N stands for network.
- A mnemonic to remember the network topologies is **Bus Rides Stink Mostly**, where B stands for bus, R stands for ring, S stands for star, and M stands for mesh.
- A mnemonic to remember the network architectures is **People Can Have Networks**, where P stands for P2P, C stands for client-server, H stands for hybrid, and N stands for network.

- A mnemonic to remember the network protocols is **The Happy Fish Swims Deep**, where T stands for TCP/IP, H stands for HTTP, F stands for FTP, S stands for SMTP, and D stands for DNS.

Goals and applications of networks and protocols

- A network is a collection of devices that can communicate with each other over a shared medium, such as cables, wireless signals, or the internet.
- A protocol is a set of rules that defines how devices exchange information on a network, such as the format, timing, and error handling of messages.
- The main goals of networks and protocols are:
 - **Resource sharing:** Networks allow multiple devices to access and use the same resources, such as printers, scanners, modems, files, and applications. This reduces the cost and increases the efficiency of using these resources. For example, a group of office workers can share a common printer instead of having one for each person.
 - **High reliability:** Networks provide multiple sources of supply and backup for the data and services that devices need. This increases the availability and fault tolerance of the network. For example, if one server fails, another server can take over its tasks and prevent data loss.
 - **Greater flexibility:** Networks enable devices to connect and communicate with each other regardless of their physical location, type, or configuration. This allows users to access data and services from anywhere and anytime, and to use different devices for different purposes. For example, a user can access their email from their laptop, smartphone, or tablet.
 - **Increased productivity:** Networks facilitate the collaboration and coordination of users and devices on a network. This improves the speed and quality of work and decision making. For example, a team of workers can share and edit documents online, chat and video conference with each other, and use online tools and platforms to manage their projects.
- The applications of networks and protocols are diverse and numerous, ranging from personal to professional, and from local to global. Some examples are:
 - **The internet:** The internet is the largest and most widely used network in the world, connecting billions of devices and users across the globe. The internet uses a suite of protocols, such as TCP/IP, HTTP, SMTP, and DNS, to enable the exchange of information and services, such as web browsing, email, online shopping, social media, and streaming.

- **Local area networks (LANs):** LANs are networks that connect devices within a small geographic area, such as a home, office, or school. LANs use protocols, such as Ethernet, Wi-Fi, and Bluetooth, to enable the sharing of resources, such as files, printers, and internet access, among the devices on the network.
- **Wide area networks (WANs):** WANs are networks that connect devices across a large geographic area, such as a city, country, or continent. WANs use protocols, such as MPLS, ATM, and Frame Relay, to enable the transmission of data and services, such as voice, video, and data, over long distances and different types of media, such as cables, satellites, and radio waves.
- **Virtual private networks (VPNs):** VPNs are networks that create a secure and private connection between devices over a public network, such as the internet. VPNs use protocols, such as IPsec, SSL, and PPTP, to encrypt and authenticate the data and devices on the network. VPNs are used for various purposes, such as remote access, online security, and anonymity.

Categories of networks in computer networks

A computer network is a system of interconnected devices that can communicate and share data, resources, and applications. Computer networks can be classified based on several criteria, such as the size, the topology, the transmission medium, and the organizational intent. Based on the size, the most common types of computer networks are:

- **Personal Area Network (PAN):** A PAN is the smallest and simplest type of network. It connects devices within the range of an individual, such as a smartphone, a laptop, a printer, or a smartwatch. A PAN is usually no larger than 10 meters. A PAN can use wired or wireless technologies, such as Bluetooth, USB, or infrared. A PAN can enable personal data transfer, device synchronization, or internet access.
- **Local Area Network (LAN):** A LAN is a network that connects devices within a limited physical area, such as a home, an office, a school, or a laboratory. A LAN can use wired or wireless technologies, such as Ethernet, Wi-Fi, or fiber optics. A LAN can enable data sharing, resource sharing, or internet access. A LAN can have multiple subnetworks, such as VLANs or WLANs. A LAN can have different topologies, such as bus, ring, star, or mesh.
- **Metropolitan Area Network (MAN):** A MAN is a network that connects devices within a large geographical area, such as a city, a town, or a campus. A MAN can use wired or wireless technologies, such as cable,

DSL, or WiMAX. A MAN can enable data sharing, resource sharing, or internet access. A MAN can have multiple LANs or WANs connected to it. A MAN can have different topologies, such as ring, star, or tree.

- **Wide Area Network (WAN):** A WAN is a network that connects devices across a large geographical area, such as a country, a continent, or the world. A WAN can use wired or wireless technologies, such as telephone lines, satellite, or cellular. A WAN can enable data sharing, resource sharing, or internet access. A WAN can have multiple LANs, MANs, or other WANs connected to it. A WAN can have different topologies, such as point-to-point, point-to-multipoint, or mesh.
- **Wireless Local Area Network (WLAN):** A WLAN is a network that connects devices using wireless technologies, such as Wi-Fi, Bluetooth, or infrared. A WLAN can be a type of LAN, MAN, or WAN, depending on the size and the coverage. A WLAN can enable data sharing, resource sharing, or internet access. A WLAN can have different topologies, such as infrastructure, ad hoc, or mesh.
- **Storage Area Network (SAN):** A SAN is a network that connects storage devices, such as hard disks, tape drives, or optical disks. A SAN can use wired or wireless technologies, such as fiber channel, iSCSI, or FCoE. A SAN can enable data storage, data backup, data recovery, or data replication. A SAN can have different topologies, such as point-to-point, arbitrated loop, or switched fabric.
- **Virtual Private Network (VPN):** A VPN is a network that connects devices using encryption and tunneling techniques. A VPN can use wired or wireless technologies, such as IPsec, SSL, or PPTP. A VPN can enable secure data transfer, remote access, or network extension. A VPN can have different topologies, such as site-to-site, client-to-server, or peer-to-peer.

Some mnemonics and learning tricks for the categories of networks in computer networks are:

- To remember the order of the network types based on the size, use the acronym **P**lease **L**et **M**e **W**atch **W**onderful **S**hows **V**ery **P**eacefully (**PAN**, **LAN**, **MAN**, **WAN**, **WLAN**, **SAN**, **VPN**).
- To remember

Organization of the Internet

- The Internet is a global network of interconnected computers and devices that communicate using standardized protocols and procedures .
- The Internet is composed of many smaller networks, called internets, that are connected by routers and gateways .

- The Internet is organized at four levels: hardware, access, navigation, and community.
 - Hardware: The physical devices and infrastructure that make up the Internet, such as computers, servers, routers, cables, satellites, etc. The hardware is owned and operated by various entities, such as governments, corporations, universities, etc.
 - Access: The services and providers that enable users to connect to the Internet, such as Internet service providers (ISPs), wireless carriers, broadband providers, etc. The access level is regulated by various laws and policies, such as net neutrality, privacy, security, etc.
 - Navigation: The tools and systems that help users find and access information and resources on the Internet, such as domain names, IP addresses, web browsers, search engines, etc. The navigation level is governed by various standards and organizations, such as the Internet Corporation for Assigned Names and Numbers (ICANN), the World Wide Web Consortium (W3C), etc.
 - Community: The sites and platforms that host and facilitate online interactions and content creation, such as social media, blogs, forums, wikis, etc. The community level is shaped by the norms and values of the users and moderators, such as terms of service, codes of conduct, etc.
- The Internet is not controlled by any single authority or organization, but rather by a distributed and collaborative network of stakeholders, such as users, developers, operators, regulators, etc. The Internet is influenced by various factors, such as technology, innovation, politics, culture, etc.
- The Internet is constantly evolving and expanding, as new devices, applications, and services are added and updated. The Internet faces various challenges and opportunities, such as security, privacy, accessibility, diversity, etc.

: **Internet Organization - freesoft.org** The organization of the Internet - Oxford University Press

ISP

- ISP stands for **Internet Service Provider** .
- An ISP is a company that provides **access to the internet** to both personal and business customers .
- ISPs make it possible for their customers to **surf the web, shop online, conduct business, and connect with family and friends**—all for a fee.
- ISPs may also provide other services such as **email services, domain registration, web hosting, and browser services** .
- ISPs can host websites for businesses and can also build the websites themselves.
- ISPs are classified into different types based on the **technology** they use

to deliver internet access, such as **dial-up, DSL, cable, fiber, satellite, wireless, mobile, etc.**

- ISPs are also classified into different tiers based on the **scope** of their network and the **number** of customers they serve, such as **Tier 1, Tier 2, Tier 3, etc.**
- Tier 1 ISPs are the **largest** and have **global** coverage. They have **direct access** to the internet backbone and do not pay any fees to other ISPs. Examples of Tier 1 ISPs are **AT&T, Verizon, CenturyLink, etc.**
- Tier 2 ISPs are **smaller** and have **regional** coverage. They have **indirect access** to the internet backbone and pay fees to Tier 1 ISPs. Examples of Tier 2 ISPs are **Comcast, Cox, Charter, etc.**
- Tier 3 ISPs are the **smallest** and have **local** coverage. They have **no access** to the internet backbone and pay fees to Tier 1 and Tier 2 ISPs. Examples of Tier 3 ISPs are **local cable companies, wireless providers, etc.**
- A mnemonic to remember the types of ISPs is **Dial-up, DSL, Cable, Fiber, Satellite, Wireless, Mobile**. A possible sentence is **Did David Call Fiona's Sister Without Mentioning**.
- A mnemonic to remember the tiers of ISPs is **Tier 1, Tier 2, Tier 3**. A possible sentence is **The Internet's Top Three**.

Network structure with reference to Computer Networks

- A computer network is a structure that makes available to a data processing user at one place some data processing function or service performed at another place.
- Computer network structure defines how computers are organized in the network and what tasks are assigned to those computers.
- Computer network structure consists of the following components:
 - Hardware: The physical devices that make up the network, such as computers, routers, switches, cables, etc.
 - Software: The programs that run on the hardware and enable communication, such as operating systems, applications, protocols, etc.
 - Transmission media: The medium that carries the data signals between the hardware devices, such as copper wires, optical fibers, radio waves, etc.
 - Network topology: The shape or layout of the network, such as bus, star, ring, mesh, etc.
 - Communications protocols: The rules and standards that govern how data is formatted, transmitted, and received on the network, such as TCP/IP, Ethernet, Wi-Fi, etc.
- Computer network structure can be classified into different types based on the size, scope, and purpose of the network, such as:
 - LAN (local area network): A network that connects computers over a relatively short distance, such as within a building or a campus.
 - WLAN (wireless local area network): A network that uses wireless

transmission media, such as radio waves or infrared, to connect computers without cables.

- WAN (wide area network): A network that connects computers over a large geographic area, such as across cities, countries, or continents.
 - MAN (metropolitan area network): A network that connects computers within a metropolitan area, such as a city or a town.
 - PAN (personal area network): A network that connects personal devices, such as smartphones, tablets, laptops, etc., usually over a short range, such as within a few meters.
 - SAN (storage area network): A network that connects storage devices, such as hard disks, tapes, etc., to provide high-speed data access and backup.
 - CAN (campus area network): A network that connects LANs within a campus or a complex, such as a university or a hospital.
 - VPN (virtual private network): A network that uses encryption and tunneling techniques to create a secure and private connection over a public network, such as the Internet.
- Computer network structure follows a layered model that defines the functions and responsibilities of each layer in the network communication process, such as:
 - Network access layer: Defines how the data gets physically transferred over the transmission media.
 - Internet layer: Packages the data into understandable packets so it can be sent and received over the network.
 - Transport layer: Allows the network devices to maintain conversations and ensure reliable and error-free data delivery.
 - Application layer: Establishes how high-level applications access the network for purposes of data transfer, such as web browsing, email, file sharing, etc.

Network architecture with reference to Computer Networks

- Network architecture is the design of a computer network .
- It is a framework for the specification of a network's physical components and their functional organization and configuration, its operational principles and procedures, as well as communication protocols used .
- Network architecture components include hardware, software, transmission media (wired or wireless), network topology, and communications protocols .
- Network architecture can be classified into two types: client-server and peer-to-peer .
 - Client-server architecture: A client-server architecture consists of one or more servers that provide services to multiple clients. The clients request services from the servers and the servers respond to the requests. The servers are usually more powerful and reliable than the clients. Examples of client-server architecture are web browsing,

email, and online gaming .

- Peer-to-peer architecture: A peer-to-peer architecture consists of a network of nodes that can act as both clients and servers. The nodes communicate directly with each other without relying on a central server. The nodes are usually equal in terms of power and responsibility. Examples of peer-to-peer architecture are file sharing, video conferencing, and blockchain .
- Network architecture can also be classified into different layers according to the functions and protocols involved. The most common layering model is the OSI (Open Systems Interconnection) model, which consists of seven layers: physical, data link, network, transport, session, presentation, and application .
 - Physical layer: The physical layer is responsible for transmitting and receiving raw bits over a physical medium, such as copper wire, optical fiber, or radio waves. It defines the characteristics of the medium, such as voltage levels, modulation schemes, and connectors .
 - Data link layer: The data link layer is responsible for organizing bits into frames and ensuring reliable and error-free transmission over a single link. It defines the addressing, framing, error detection, and flow control mechanisms. Examples of data link layer protocols are Ethernet, Wi-Fi, and PPP .
 - Network layer: The network layer is responsible for routing packets across multiple links and networks. It defines the addressing, routing, forwarding, and congestion control mechanisms. Examples of network layer protocols are IP, ICMP, and ARP .
 - Transport layer: The transport layer is responsible for providing end-to-end communication between applications. It defines the port numbers, segmentation, reassembly, reliability, and flow control mechanisms. Examples of transport layer protocols are TCP, UDP, and SCTP .
 - Session layer: The session layer is responsible for establishing, maintaining, and terminating sessions between applications. It defines the synchronization, checkpointing, and recovery mechanisms. Examples of session layer protocols are RPC, NFS, and SQL .
 - Presentation layer: The presentation layer is responsible for translating, encrypting, and compressing data between applications. It defines the data formats, encryption schemes, and compression algorithms. Examples of presentation layer protocols are SSL, TLS, and JPEG .
 - Application layer: The application layer is responsible for providing specific services to users. It defines the user interfaces, commands, and messages. Examples of application layer protocols are HTTP, FTP, and SMTP .
- A mnemonic to remember the OSI layers from bottom to top is “Please Do Not Throw Sausage Pizza Away” or “All People Seem To Need Data Processing” .

- A network architecture diagram is a graphical representation of the components and connections of a network. It can help to visualize the structure, function, and performance of a network. An example of a network architecture diagram is shown below:

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Layering principles with reference to Network Architecture in Computer Networks

- Layering is a design principle that divides a complex system into smaller and simpler components, called layers, that can be managed independently.
- Each layer has a specific function and provides services to the layer above it, while using the services of the layer below it.
- Layering allows for modularity, interoperability, scalability, and flexibility of network systems, as well as hiding the implementation details of each layer from other layers.
- The most widely used layered network architecture is the Open Systems Interconnection (OSI) model, which consists of seven layers: physical, data link, network, transport, session, presentation, and application.
- The OSI model defines the functions, protocols, and interfaces of each layer, and serves as a reference for developing and standardizing network protocols and devices.
- The following table summarizes the main features and examples of each layer of the OSI model:

Layer	Function	Protocols	Devices
Physical	Transmits raw bits over a physical medium	Ethernet, RS-232, Bluetooth	Hubs, repeaters, cables
Data link	Provides reliable and error-free transmission of frames between adjacent nodes	HDLC, PPP, MAC, LLC	Switches, bridges, NICs
Network	Provides logical addressing and routing of packets across networks	IP, ICMP, ARP, RIP, OSPF	Routers, firewalls, gateways
Transport	Provides end-to-end reliable and ordered delivery of data segments	TCP, UDP, SCTP	N/A

Layer	Function	Protocols	Devices
Session	Establishes, maintains, and terminates sessions between applications	RPC, NFS, SQL, NetBIOS	N/A
Presentation	Translates, encrypts, and compresses data for the application layer	SSL, TLS, JPEG, GIF	N/A
Application	Provides user interface and high-level services for network applications	HTTP, FTP, SMTP, DNS, Telnet	Web browsers, email clients, servers

- A possible mnemonic to remember the order of the OSI layers from bottom to top is: **P**lease **D**o **N**ot **T**hrow **S**ausage **P**izza **A**way.

Services in networks architecture in Computer Networks

- Services in networks architecture are the **functions** or **operations** that each layer of the network model provides to the layer above it or below it.
- Services are **abstract** and **independent** of the implementation details of the layer that provides them.
- Services can be **connection-oriented** or **connectionless**, depending on whether they require a logical link between the sender and the receiver or not.
- Services can be **reliable** or **unreliable**, depending on whether they guarantee the delivery of data or not.
- Services can be **synchronous** or **asynchronous**, depending on whether they require the sender and the receiver to be synchronized or not.
- Services can be **request-response** or **indication-confirmation**, depending on whether they involve a single or a double exchange of messages between the sender and the receiver.
- Services can be **unicast**, **multicast**, **broadcast**, or **anycast**, depending on whether they deliver data to one, many, all, or any destination(s) respectively.
- Services can be **stateful** or **stateless**, depending on whether they maintain information about the state of the communication or not.
- Services can be **peer-to-peer** or **client-server**, depending on whether they involve equal or unequal roles of the communicating parties.

Some examples of services in networks architecture are:

- **DHCP (Dynamic Host Configuration Protocol):** A service that assigns IP addresses and other network configuration parameters to devices on a network .
- **DNS (Domain Name System):** A service that translates domain names into IP addresses and vice versa .
- **FTP (File Transfer Protocol):** A service that allows users to transfer files between hosts on a network.
- **HTTP (Hypertext Transfer Protocol):** A service that allows users to access web pages and other resources on a network.
- **SMTP (Simple Mail Transfer Protocol):** A service that allows users to send and receive email messages on a network.
- **SNMP (Simple Network Management Protocol):** A service that allows network administrators to monitor and manage network devices and services.

A possible mnemonic to remember some of the services in networks architecture is:

Don't Do Foolish Habits Such Smoking Nicotine More Per Hour

where each letter stands for:

- **DHCP**
- **DNS**
- **FTP**
- **HTTP**
- **SMTP**
- **SNMP**
- **Network**
- **Management**
- **Protocol**
- **Hour**

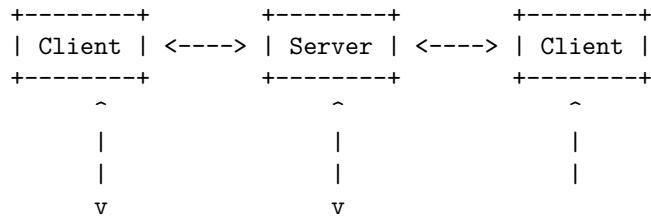
protocols and standards in networks architecture in Computer Networks

- Network protocols are the rules and procedures that define how devices communicate and exchange data over a network .
- Network standards are the specifications and guidelines that ensure compatibility and interoperability among different network devices, systems, and applications .
- Network architecture is the design and structure of a network, including its components, protocols, standards, and topology.
- Some of the common network protocols and standards are:

- TCP/IP: Transmission Control Protocol/Internet Protocol is the most widely used protocol suite that provides end-to-end data transmission and addressing over the Internet and other networks .
 - UDP: User Datagram Protocol is a connectionless protocol that provides fast and unreliable data delivery over the Internet and other networks .
 - ARP: Address Resolution Protocol is a protocol that maps network layer addresses (such as IP addresses) to data link layer addresses (such as MAC addresses) on a local area network (LAN) .
 - DHCP: Dynamic Host Configuration Protocol is a protocol that assigns network configuration parameters (such as IP addresses, subnet masks, and default gateways) to devices on a network dynamically .
 - FTP: File Transfer Protocol is a protocol that allows users to transfer files between devices over a network .
 - DNS: Domain Name System is a protocol that translates domain names (such as www.google.com) to IP addresses (such as 172.217.14.206) on the Internet and other networks .
 - HTTP: Hypertext Transfer Protocol is a protocol that defines how web browsers and web servers communicate and exchange web pages and other resources over the Internet and other networks .
 - SMTP: Simple Mail Transfer Protocol is a protocol that defines how email messages are sent and received over the Internet and other networks .
 - IEEE 802: A family of standards that define various aspects of data link layer and physical layer protocols for different types of networks, such as Ethernet, Wi-Fi, Bluetooth, and ZigBee.
 - OSI: Open Systems Interconnection is a reference model that defines seven layers of network protocols and functions, from the physical layer to the application layer.
- A mnemonic to remember the seven layers of the OSI model is: **Please Do Not Throw Sausage Pizza Away.**
 - Physical layer: deals with the transmission and reception of raw bits over a physical medium, such as cables, wires, or radio waves.
 - Data link layer: deals with the framing, error detection, and flow control of data packets over a physical medium, such as Ethernet or Wi-Fi.
 - Network layer: deals with the routing, addressing, and fragmentation of data packets over a network, such as IP or ICMP.
 - Transport layer: deals with the end-to-end delivery, reliability, and congestion control of data segments over a network, such as TCP or UDP.
 - Session layer: deals with the establishment, maintenance, and termination of sessions between applications over a network, such as RPC or SSL.
 - Presentation layer: deals with the representation, encryption, and

compression of data between applications over a network, such as ASCII or JPEG.

- Application layer: deals with the user interface, functionality, and protocols of applications over a network, such as HTTP or FTP.
- An example of network architecture is the client/server model, where one or more devices (clients) request and receive services from another device (server) over a network.
 - A client is a device that initiates a communication and requests a service from a server, such as a web browser or an email client.
 - A server is a device that responds to a communication and provides a service to a client, such as a web server or an email server.
 - A network is a collection of devices that are connected by communication links and protocols, such as the Internet or a LAN.
 - A client/server network architecture can be illustrated as follows:



The OSI reference model in in Computer Networks

- The OSI reference model stands for Open Systems Interconnection reference model. It is a conceptual framework that defines how different network devices and software applications communicate with each other across a network.
- The OSI reference model consists of seven layers, each of which performs a specific function in the process of data transmission. The layers are:
 - **Physical layer:** This layer deals with the physical aspects of data transmission, such as the electrical signals, cables, connectors, and hardware devices. It defines the standards for encoding, modulation, synchronization, and error detection and correction of data bits.
 - **Data link layer:** This layer provides reliable and efficient data transfer between two adjacent nodes on a network. It defines the protocols for framing, addressing, flow control, and error control of data packets. It also handles the access and allocation of the shared medium among multiple nodes.
 - **Network layer:** This layer is responsible for routing and forwarding data packets across different networks. It defines the protocols for addressing, routing, congestion control, and internetworking of data packets. It also handles the fragmentation and reassembly of data

packets when they are too large for the underlying network.

- **Transport layer:** This layer provides end-to-end data delivery between two applications on a network. It defines the protocols for establishing, maintaining, and terminating connections, as well as for ensuring the reliability, integrity, and quality of service of data transmission. It also handles the segmentation and reassembly of data streams when they are too large for the underlying network.
 - **Session layer:** This layer manages the communication sessions between two applications on a network. It defines the protocols for initiating, controlling, and terminating sessions, as well as for synchronizing, coordinating, and recovering data exchange within a session. It also handles the authentication and authorization of users and applications.
 - **Presentation layer:** This layer handles the representation and transformation of data between different formats and encodings. It defines the protocols for data compression, encryption, decryption, translation, and conversion of data. It also handles the abstraction and interpretation of data for different applications.
 - **Application layer:** This layer provides the interface and functionality for various network applications and services. It defines the protocols for data exchange, remote access, file transfer, email, web browsing, and other network services. It also handles the user interaction and presentation of data for different applications.
- The OSI reference model is useful for understanding, designing, and troubleshooting network systems and applications. It provides a common language and framework for describing the functions and interactions of different network components. It also allows interoperability and compatibility among different network devices and software applications that follow the same standards and protocols.
 - A mnemonic to remember the names and order of the OSI layers is: **Please Do Not Throw Sausage Pizza Away**. Another mnemonic is: **All People Seem To Need Data Processing**.

TCP/IP protocol suite in in Computer Networks

- TCP/IP stands for Transmission Control Protocol/Internet Protocol and is a suite of communication protocols used to interconnect network devices on the internet or other computer networks.
- TCP/IP is also known as the internet protocol suite, as it defines the basic protocols for the internet, such as IP, TCP, UDP, ICMP, ARP, etc .
- TCP/IP is based on a four-layer model, which consists of the following layers :
 - Application layer: This layer provides the interface for the user applications to communicate with the network, such as HTTP, FTP, SMTP, DNS, etc.

- Transport layer: This layer provides reliable or unreliable data delivery services between the end hosts, such as TCP, UDP, etc.
- Internet layer: This layer provides the logical addressing and routing of packets across the network, such as IP, ICMP, ARP, etc.
- Network access layer: This layer provides the physical and data link functions to transmit and receive bits on the network medium, such as Ethernet, Wi-Fi, PPP, etc.
- TCP/IP is a standard protocol suite that is widely used and supported by various operating systems, devices, and applications.
- TCP/IP is a flexible and scalable protocol suite that can adapt to different network architectures and technologies.
- TCP/IP is a robust and fault-tolerant protocol suite that can handle network failures and congestion.

Some mnemonics and learning tricks for TCP/IP protocol suite are:

- To remember the four layers of TCP/IP model, use the acronym **ATIN** (Application, Transport, Internet, Network access) or **NITA** (Network access, Internet, Transport, Application).
- To remember some of the common protocols in each layer of TCP/IP model, use the following phrases:
 - Application layer: **How Fast Should Data Travel?** (**H**TTP, **F**TP, **S**MTP, **D**NS, **T**ELNET)
 - Transport layer: **The Ultimate Data Gram** (**T**CP, **U**DP, **D**CCP, **G**TP)
 - Internet layer: **I Can Always Reach People** (**I**P, **I**CMP, **I**GMP, **A**RP, **R**ARP, **P**IM)
 - Network access layer: **Everyone Wants Pretty Slippers** (**E**thernet, **W**i-Fi, **P**PP, **S**LIP)

Network devices in Computer Networks

Network devices are physical devices that enable communication and interaction between hardware on a computer network. Each networking device operates in a distinct computer network segment and performs distinct functions. A network may require hundreds or thousands of different network devices to maintain and build out various LAN and WAN.

Some of the common types of network devices are:

- **Repeater:** A repeater is a device that operates at the physical layer of the OSI model. Its job is to regenerate the signal over the same network before it becomes weak or corrupted due to attenuation. A repeater can extend the transmission distance of a network segment, but it cannot filter or segment the traffic.
- **Hub:** A hub is a device that operates at the physical layer of the OSI model. It is a multiport repeater that connects multiple wires coming

from different branches. A hub broadcasts the data received from one port to all other ports. A hub can increase the number of devices on a network, but it cannot reduce the network congestion or collision.

- **Bridge:** A bridge is a device that operates at the data link layer of the OSI model. It is used to connect two or more network segments that use the same or different physical media. A bridge can filter the traffic based on the MAC addresses of the devices. A bridge can reduce the network congestion and collision by dividing the network into smaller segments.
- **Switch:** A switch is a device that operates at the data link layer or the network layer of the OSI model. It is a multiport bridge that can store and forward the data frames based on the MAC addresses or the IP addresses of the devices. A switch can create multiple collision domains and broadcast domains within a network. A switch can improve the network performance and security by reducing the network congestion and collision .
- **Router:** A router is a device that operates at the network layer of the OSI model. It is used to connect two or more networks that use the same or different network protocols. A router can route the data packets based on the IP addresses of the devices. A router can create multiple broadcast domains within a network. A router can enable the communication between different networks and support various network services such as NAT, DHCP, VPN, etc .
- **Gateway:** A gateway is a device that operates at the application layer of the OSI model. It is used to connect two or more networks that use different network architectures or protocols. A gateway can translate the data formats and protocols between different networks. A gateway can enable the communication between heterogeneous networks and support various network applications such as email, web, FTP, etc.
- **Brouter:** A brouter is a device that combines the functions of a bridge and a router. It can operate at both the data link layer and the network layer of the OSI model. It can filter and route the data packets based on the MAC addresses or the IP addresses of the devices. A brouter can provide the benefits of both a bridge and a router, such as reducing the network congestion and collision, creating multiple broadcast domains, and enabling the communication between different networks.
- **NIC:** A NIC (Network Interface Card) is a device that operates at the physical layer and the data link layer of the OSI model. It is a hardware component that connects a computer or a device to a network. A NIC can encode and decode the data signals and provide the MAC address for the device. A NIC can support various network standards and speeds, such as Ethernet, Wi-Fi, Bluetooth, etc.

A possible mnemonic to remember the types of network devices is:

Repeat **H**ub **B**ridge **S**witch **R**outer **G**ateway **B**router **N**IC

or

Really Helpful Boys Share Rare Gems Bravely Now <https://www.scaler.com/topics/computer-network/network-devices/>

<https://www.geeksforgeeks.org/network-devices-hub-repeater-bridge-switch-router-gateways/>

<https://testbook.com/learn/types-of-computer-network-devices/>

Network components in Computer Networks

Network components are the devices and software that enable communication and data exchange between computers and other networked devices. Some of the major network components are:

- **Network Interface Card (NIC):** A NIC is a hardware device that allows a computer to connect to a network. It provides a physical interface for sending and receiving data over a network medium, such as a cable or wireless signal. A NIC has a unique address, called a MAC address, that identifies the device on the network .
- **Hub:** A hub is a device that connects multiple computers or other network devices together. It acts as a central point for data transmission, and broadcasts any data it receives to all the connected devices. A hub operates at the physical layer of the network, and does not perform any filtering or routing of data. Hubs are simple and cheap, but they can cause network congestion and collisions .
- **Switch:** A switch is a device that connects multiple computers or other network devices together. It acts as a central point for data transmission, and forwards data only to the intended destination device. A switch operates at the data link layer of the network, and uses MAC addresses to identify the devices on the network. Switches are more intelligent and efficient than hubs, but they are also more expensive .
- **Cables and connectors:** Cables and connectors are the physical components that link the network devices together. They provide the medium for data transmission, and can be of different types, such as twisted pair, coaxial, fiber optic, etc. Connectors are the devices that attach the cables to the network devices, and can be of different types, such as RJ-45, BNC, SC, etc. Cables and connectors have different specifications and standards, such as speed, bandwidth, distance, etc .
- **Router:** A router is a device that connects multiple networks together. It acts as a gateway for data transmission, and routes data to the best path based on the network address of the destination device. A router operates at the network layer of the network, and uses IP addresses to identify the devices on the network. Routers are more complex and powerful than switches, but they are also more costly and require more configuration .
- **Modem:** A modem is a device that modulates and demodulates analog signals into digital signals, and vice versa. It enables data transmission

over different types of networks, such as telephone lines, cable lines, satellite links, etc. A modem operates at the physical layer of the network, and can have different speeds and standards, such as dial-up, DSL, cable, etc .

- **Server:** A server is a computer that provides services or resources to other computers or devices on the network. It can perform various functions, such as file sharing, web hosting, email, database, etc. A server operates at the application layer of the network, and can have different protocols and software, such as HTTP, FTP, SMTP, SQL, etc .
- **Client:** A client is a computer or device that requests services or resources from a server on the network. It can perform various functions, such as browsing, downloading, sending, querying, etc. A client operates at the application layer of the network, and can have different protocols and software, such as HTTP, FTP, SMTP, SQL, etc .
- **Firewall:** A firewall is a device or software that monitors and controls the incoming and outgoing network traffic. It acts as a barrier between the internal and external networks, and blocks or allows data based on predefined rules or policies. A firewall can operate at different layers of the network, and can have different types, such as packet filtering, stateful inspection, proxy, etc .
- **Access point:** An access point is a device that enables wireless communication between the wired and wireless network devices. It acts as a bridge between the two types of networks, and converts the signals from one medium to another. An access point operates at the physical and data link layers of the network, and can have different standards and frequencies, such as Wi-Fi, Bluetooth, etc .

Some mnemonics and learning tricks for the network components are:

- To remember the order of the network layers, use the phrase ****Please Do Not Throw Sausage Pizza**

Physical Layer in Computer Networks

- The physical layer is the lowest and first layer of the OSI model of computer networking.
- The physical layer provides an electrical, mechanical, and procedural interface to the transmission medium.
- The physical layer defines the means of transmitting a stream of raw bits over a physical data link connecting network nodes.
- The physical layer consists of the electronic circuit transmission technologies of a network, such as wires, cables, connectors, transmitters, receivers, modems, etc.
- The physical layer performs the following functions:
 - It maintains the data rate (how many bits a sender can send per second).
 - It performs synchronization of bits (how the sender and receiver agree

- on when a bit starts and ends).
 - It helps in transmission medium decision (direction of data transfer, such as simplex, half-duplex, or full-duplex).
 - It helps in physical topology decision (shape of the network, such as mesh, star, bus, ring, etc).
 - It helps in encoding and decoding of bits (how the bits are represented as physical signals, such as voltage levels, light pulses, etc).
 - It helps in multiplexing and demultiplexing of signals (how multiple signals are combined or separated on a single medium, such as frequency division multiplexing, time division multiplexing, etc).
 - It helps in error detection and correction (how the sender and receiver detect and correct errors in the bitstream, such as parity check, checksum, cyclic redundancy check, etc).
- A mnemonic to remember the functions of the physical layer is **D-SPEED**:
 - **D**ata rate
 - **S**ynchronization
 - **P**hysical medium
 - **E**ncoding
 - **E**rror detection
 - **D**uplexing
- Some examples of physical layer standards and protocols are:
 - Ethernet
 - Wi-Fi
 - Bluetooth
 - USB
 - RS-232
 - SONET
 - DSL
 - ISDN
 - GSM
 - CDMA
- Some advantages of the physical layer are:
 - It enables the transmission of data over different types of media and distances.
 - It supports various data rates and modes of communication.
 - It provides error detection and correction mechanisms to ensure reliable data transfer.
 - It allows multiplexing of multiple signals on a single medium to increase efficiency and bandwidth utilization.
- Some disadvantages of the physical layer are:
 - It does not provide any security or encryption of data.
 - It does not guarantee the delivery or order of data.
 - It does not handle congestion or flow control of data.
 - It does not provide any logical addressing or routing of data.
- Some applications of the physical layer are:

- It enables the communication between different devices and networks.
- It supports various types of services and applications, such as voice, video, data, etc.
- It facilitates the interoperability and compatibility of different standards and protocols.

Network topology design in Computer Networks

- Network topology is used to describe the physical and logical structure of a network.
- Physical topology refers to the actual layout of the computer cables and other network devices.
- Logical topology refers to the way data flows between the network devices.
- Network topology affects the performance, reliability, scalability, and security of a network.
- Network topology design is the process of selecting the most appropriate topology for a given network based on the following factors:
 - The number and location of the network nodes (such as computers, routers, switches, etc.)
 - The bandwidth and latency requirements of the network applications
 - The cost and availability of the network resources (such as cables, connectors, hubs, etc.)
 - The security and redundancy needs of the network
 - The ease of installation, maintenance, and troubleshooting of the network
- There are several types of network topologies, each with its own advantages and disadvantages. Some of the common network topologies are:
 - Bus topology: A linear arrangement of nodes connected by a single cable. It is simple, cheap, and easy to install, but it has low bandwidth, high collision rate, and poor scalability and fault tolerance.
 - Ring topology: A circular arrangement of nodes connected by a single cable. It has higher bandwidth and lower collision rate than bus topology, but it has high latency, complex installation and maintenance, and poor fault tolerance.
 - Star topology: A central node (such as a hub or a switch) connects to multiple peripheral nodes by individual cables. It has high bandwidth, low latency, and easy installation and maintenance, but it has high cost, high dependency on the central node, and poor scalability.
 - Mesh topology: A network where each node is connected to every other node by a dedicated link. It has high bandwidth, low latency, high fault tolerance, and high security, but it has very high cost, complex installation and maintenance, and high network overhead.
 - Tree topology: A hierarchical arrangement of nodes where each level is a star topology. It has high scalability, easy management, and moderate cost, but it has high dependency on the root node, high

latency, and moderate fault tolerance.

Types of connections in Computer Networks

A connection in a computer network is a link between two or more devices that allows them to communicate and share data, resources, and applications. There are different types of connections in computer networks, depending on the following factors:

- The number of devices involved in the connection
- The direction of data flow in the connection
- The physical medium used to establish the connection
- The logical topology of the connection
- The geographical scope of the connection

Based on these factors, the following are some of the common types of connections in computer networks:

- **Point-to-point connection:** This type of connection allows one device to communicate with one other device. For example, two phones may pair with each other using Bluetooth to exchange files or make calls. A point-to-point connection can be either simplex (one-way), half-duplex (two-way but not at the same time), or full-duplex (two-way and simultaneous). A point-to-point connection can use either wired or wireless medium, and can have different logical topologies, such as bus, ring, or star. A point-to-point connection can span across different geographical areas, such as local, metropolitan, or wide.
- **Broadcast/multicast connection:** This type of connection allows a device to send one message out to the network and have copies of that message delivered to multiple recipients. For example, a radio station may broadcast its signal to many listeners, or a video conferencing application may multicast its stream to many participants. A broadcast/multicast connection is usually simplex, and can use either wired or wireless medium, such as Ethernet, Wi-Fi, or satellite. A broadcast/multicast connection can have different logical topologies, such as bus, ring, or tree. A broadcast/multicast connection can span across different geographical areas, such as local, metropolitan, or wide.
- **Multipoint connection:** This type of connection allows one device to connect and deliver messages to multiple devices in parallel. For example, a router may connect to several computers and forward packets to them, or a printer may connect to several computers and print documents from them. A multipoint connection can be either simplex, half-duplex, or full-duplex, and can use either wired or wireless medium, such as USB, Bluetooth, or Wi-Fi. A multipoint connection can have different logical topologies, such as star, mesh, or hybrid. A multipoint connection can span across different geographical areas, such as local, metropolitan, or

wide.

Some mnemonics and learning tricks for the types of connections in computer networks are:

- To remember the difference between point-to-point and multipoint connections, think of a point as a device and a multipoint as a hub or a switch that connects multiple devices.
- To remember the difference between broadcast and multicast connections, think of a broadcast as a one-to-many communication and a multicast as a one-to-some communication.
- To remember the difference between simplex, half-duplex, and full-duplex connections, think of a simplex as a one-way street, a half-duplex as a two-way street with traffic lights, and a full-duplex as a two-way street with separate lanes.

Transmission media in Computer Networks

- Transmission media in computer networks are the physical channels that carry data from one device to another using electromagnetic signals .
- Transmission media can be classified into two main types: guided media and unguided media.
- Guided media are those that provide a physical path for the signals, such as wires, cables, or optical fibers. Examples of guided media are twisted pair, coaxial cable, and fiber optic cable .
- Unguided media are those that do not require a physical path for the signals, such as air, water, or vacuum. Examples of unguided media are radio waves, microwaves, infrared, and visible light .
- The choice of transmission media depends on several factors, such as bandwidth, cost, distance, reliability, security, and interference .
- Bandwidth is the amount of data that can be transmitted per unit time over a medium. Higher bandwidth means faster data transmission and higher quality of service.
- Cost is the amount of money required to install, maintain, and operate a medium. Lower cost means more affordability and accessibility.
- Distance is the maximum length of a medium that can support data transmission without significant loss or distortion. Longer distance means more coverage and scalability.
- Reliability is the degree of consistency and accuracy of data transmission over a medium. Higher reliability means less errors and retransmissions.
- Security is the level of protection of data transmission from unauthorized

access, modification, or interception. Higher security means more confidentiality and integrity.

- Interference is the unwanted noise or signals that affect the quality of data transmission over a medium. Lower interference means more clarity and stability.
- A mnemonic to remember the types of transmission media is **GURU**: Guided, Unguided, Radio, and Ultraviolet.
- A mnemonic to remember the factors that affect the choice of transmission media is **BC DRIS**: Bandwidth, Cost, Distance, Reliability, Interference, and Security.

Signal transmission and encoding in Computer Networks

- Signal transmission is the process of sending digital or analog data over a physical medium such as a wire, a cable, or a wireless channel.
- Encoding is the process of transforming data into a format that can be transmitted and understood by the receiver.
- There are different types of encoding techniques depending on the nature of the data and the medium.
- Some common encoding techniques are:
 - Digital-to-digital encoding: This is the representation of digital information by a digital signal. For example, converting binary 1s and 0s created by a computer into a series of voltage pulses that can be transmitted over a wire. This is also known as line coding. Some examples of digital-to-digital encoding are:
 - * Non-return-to-zero (NRZ): This is the simplest form of encoding, where a high voltage represents a 1 and a low voltage represents a 0. There is no change in the signal level for consecutive 1s or 0s. This encoding has the advantage of being easy to implement and having a low bandwidth requirement. However, it has the disadvantage of having no synchronization mechanism, which means that the receiver may lose track of the bit boundaries if the signal is corrupted or distorted. It also has a DC component, which means that the average voltage is not zero, which may cause problems for some transmission media.
 - * Return-to-zero (RZ): This is a variation of NRZ, where the signal level returns to zero after each bit. This encoding has the advantage of having a synchronization mechanism, which means that the receiver can detect the bit boundaries by the transitions in the signal. It also has no DC component, which means that the

average voltage is zero, which may be suitable for some transmission media. However, it has the disadvantage of having a higher bandwidth requirement, as it uses two signal levels for each bit.

- * Manchester encoding: This is a type of bipolar encoding, where the signal level changes from high to low or from low to high in the middle of each bit. This encoding has the advantage of having a synchronization mechanism, as the receiver can detect the bit boundaries by the transitions in the signal. It also has no DC component, as the average voltage is zero. However, it has the disadvantage of having a higher bandwidth requirement, as it uses two transitions for each bit.
- * Differential Manchester encoding: This is a variation of Manchester encoding, where the signal level changes in the middle of each bit only if the bit is 0. If the bit is 1, the signal level does not change in the middle of the bit, but it changes at the beginning of the next bit. This encoding has the advantage of being more robust to noise and distortion, as the receiver can detect the bit values by the presence or absence of transitions in the signal. It also has no DC component, as the average voltage is zero. However, it has the disadvantage of having a higher bandwidth requirement, as it uses two transitions for each bit.
- Analog-to-digital encoding: This is the representation of analog information by a digital signal. For example, converting a voice signal into a series of binary digits that can be transmitted over a digital network. This is also known as digitization or sampling. The process of analog-to-digital encoding involves three steps:
 - * Sampling: This is the process of taking periodic snapshots of the analog signal at a fixed rate, called the sampling rate. The sampling rate determines the quality and the bandwidth of the digital signal. According to the Nyquist theorem, the sampling rate must be at least twice the highest frequency component of the analog signal to avoid aliasing, which is the distortion of the signal due to insufficient sampling.
 - * Quantization: This is the process of assigning a discrete value to each sample, based on its amplitude. The quantization level determines the accuracy and the resolution of the digital signal. The quantization level is usually expressed in terms of the number of bits per sample, which determines the range of possible values for each sample. For example, if the quantization level is 8 bits per sample, then each sample can have $2^8 = 256$ possible values.
 - * Encoding: This is the process of converting each quantized sample into a binary code, using a predefined mapping scheme. The

encoding scheme determines the format and the size of the digital signal. For example, if the encoding scheme is pulse code modulation (PCM), then each sample is represented by a fixed-length binary code, which is then transmitted as a series of voltage pulses.

- Digital-to-analog encoding: This is the representation of digital information by an

Network performance and transmission impairments in Computer Networks

- Network performance is the measure of how well a network can deliver data and services to its users. It can be evaluated by various metrics, such as throughput, delay, jitter, packet loss, availability, reliability, etc.
- Transmission impairments are the factors that degrade the quality of a signal as it travels through a transmission medium. They can cause errors, distortion, or loss of data in the communication process. Some common types of transmission impairments are:
 - Attenuation: The gradual weakening of a signal as it travels over a distance. It can be caused by the resistance of the medium, the absorption of the signal by the medium, or the leakage of the signal from the medium. Attenuation can be measured in decibels (dB), which indicate the ratio of the input power to the output power of a signal. A negative dB value means that the signal has lost power, while a positive dB value means that the signal has gained power (e.g., by amplification).
 - Distortion: The alteration of the shape or form of a signal as it travels through a medium. It can be caused by the non-linear characteristics of the medium, the interference of other signals, or the variation of the propagation speed of the signal in the medium. Distortion can affect the amplitude, phase, or frequency of a signal, and can result in errors or misinterpretation of the data. Distortion can be classified into two types: linear distortion and nonlinear distortion. Linear distortion occurs when the amplitude and phase of the signal are affected uniformly across the frequency spectrum, while nonlinear distortion occurs when the amplitude and phase of the signal are affected differently at different frequencies.
 - Noise: The unwanted or random signals that are added to the original signal as it travels through a medium. It can be caused by the thermal agitation of electrons in the medium, the electromagnetic interference from other sources, or the impulse or spike signals from external events. Noise can reduce the signal-to-noise ratio (SNR), which is the ratio of the power of the signal to the power of the noise. A low SNR means that the signal is hard to distinguish from the noise, while a high SNR means that the signal is clear and easy to detect. Noise can

be classified into three types: thermal noise, crosstalk, and impulse noise. Thermal noise is the random noise generated by the thermal motion of electrons in the medium, and it is proportional to the temperature and the bandwidth of the signal. Crosstalk is the noise caused by the coupling of signals from adjacent wires or channels, and it can result in interference or leakage of data. Impulse noise is the noise caused by sudden and short-duration changes in the voltage or current of the signal, and it can result from lightning, switching, or other external events.

- A mnemonic to remember the types of transmission impairments is: **A**ttenuation, **D**istortion, and **N**oise (ADN).
- A learning trick to understand the concept of attenuation is to think of a flashlight. As the flashlight moves away from a wall, the light beam becomes weaker and dimmer. This is similar to how a signal loses power as it travels over a distance.
- A learning trick to understand the concept of distortion is to think of a music player. As the music player plays a song, the sound quality may vary depending on the volume, the equalizer settings, or the interference from other devices. This is similar to how a signal changes shape or form as it travels through a medium.
- A learning trick to understand the concept of noise is to think of a radio. As the radio tunes to a station, the sound may be clear or fuzzy depending on the signal strength, the frequency, or the presence of other stations. This is similar to how a signal is affected by unwanted or random signals as it travels through a medium.

Switching techniques and multiplexing in Computer Networks

Switching techniques are methods of connecting multiple devices in a network and transferring data between them. There are three main types of switching techniques: circuit switching, message switching, and packet switching.

- **Circuit switching:** In circuit switching, two nodes communicate with each other over a dedicated communication path. The path is established before the data transmission and remains active until the communication is over. The advantage of circuit switching is that it provides a guaranteed and continuous connection between the nodes. The disadvantage is that it wastes bandwidth if the connection is idle or the data rate is low.
- **Message switching:** In message switching, the whole message is treated as a data unit. The message is stored and forwarded by intermediate nodes until it reaches the destination. The advantage of message switching is that it does not require a dedicated path and can avoid congestion by buffering the messages. The disadvantage is that it introduces delay and overhead due to the storage and forwarding process.
- **Packet switching:** The packet switching technique is derived from message switching where the message is broken down into smaller chunks called

packets. The packets are transmitted independently and may take different routes to reach the destination. The advantage of packet switching is that it can utilize the bandwidth efficiently and adapt to dynamic network conditions. The disadvantage is that it may cause packet loss, duplication, or reordering due to the network errors or congestion.

Multiplexing is a technique of combining multiple signals into one signal over a shared medium. Multiplexing allows the simultaneous transmission of multiple signals across a single data link. There are different types of multiplexing techniques, such as frequency division multiplexing (FDM), time division multiplexing (TDM), wavelength division multiplexing (WDM), and code division multiplexing (CDM).

- Frequency division multiplexing: In FDM, the bandwidth of the channel is divided into several frequency bands, and each signal is modulated by a different carrier frequency. The modulated signals are then combined and transmitted over the channel. The advantage of FDM is that it is simple and robust. The disadvantage is that it may cause interference and crosstalk between the signals.
- Time division multiplexing: In TDM, the time of the channel is divided into several time slots, and each signal is assigned a different time slot. The signals are then transmitted one after another over the channel. The advantage of TDM is that it can avoid interference and crosstalk between the signals. The disadvantage is that it may cause synchronization and framing issues.
- Wavelength division multiplexing: In WDM, the wavelength of the light is divided into several bands, and each signal is modulated by a different wavelength. The modulated signals are then combined and transmitted over the optical fiber. The advantage of WDM is that it can increase the capacity and bandwidth of the optical fiber. The disadvantage is that it may require expensive and complex devices.
- Code division multiplexing: In CDM, each signal is encoded by a different code, and the encoded signals are then combined and transmitted over the channel. The advantage of CDM is that it can provide security and privacy for the signals. The disadvantage is that it may require complex encoding and decoding algorithms.

Statistical multiplexing is a communication link sharing technique, which is used in packet switching. The shared linking is variable in statistical multiplexing, whereas it is fixed in TDM or FDM. This is a strategic application for maximizing the utilization of bandwidth. This can increase the efficiency of network, as well.

Unit 2 - Link layer in Computer Networks and Medium Access Control and Local Area Networks

- The link layer is the lowest layer in the TCP/IP network model and is equivalent to layer 2 in the OSI model.
- The link layer is responsible for transferring data between adjacent network nodes in a wide area network or between nodes on the same local area network.
- The link layer consists of two sublayers: the logical link control (LLC) sublayer and the media access control (MAC) sublayer .
- The LLC sublayer is responsible for managing communications links and handling frame traffic.
- The MAC sublayer is responsible for governing protocol access to the physical network medium.
- The MAC sublayer defines how nodes share the common channel and avoid collisions.
- The MAC sublayer can use different techniques to coordinate the access to the channel, such as contention-based, reservation-based, polling-based, or token-based.
- The MAC sublayer can also use different protocols to handle collisions, such as CSMA/CD, CSMA/CA, or ALOHA.
- The MAC sublayer can be implemented in different ways depending on the type of network, such as Ethernet, Wi-Fi, Bluetooth, or cellular.
- The link layer frames are the smallest units of bits on a link layer network and contain a header, a payload, and a trailer.
- The link layer header contains information such as the source and destination MAC addresses, the frame type, and the frame length.
- The link layer trailer contains information such as the frame check sequence, which is used to detect errors.
- The link layer can provide various services to the network layer, such as framing, error detection, error correction, flow control, and link management.
- The link layer can also provide various functions to the physical layer, such as encoding, decoding, synchronization, and modulation.

Mnemonics and learning tricks

- To remember the two sublayers of the link layer, use the acronym **Link Layer = LLC + MAC**.
- To remember the four types of MAC techniques, use the acronym **Contention, Reservation, Polling, Token = CRPT**.
- To remember the three types of collision handling protocols, use the acronym **CSMA/CD, CSMA/CA, ALOHA = CCA**.
- To remember the four types of network where MAC can be implemented, use the acronym **Ethernet, Wi-Fi, Bluetooth, Cellular = EWBC**.
- To remember the three parts of a link layer frame, use the acronym **Header,**

Payload, Trailer = **HPT**.

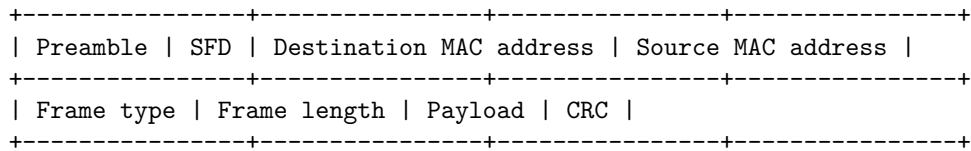
- To remember the five services that the link layer can provide to the network layer, use the acronym **F**raming, **E**rror detection, **E**rror correction, **F**low control, **L**ink management = **FEEL F**.
- To remember the four functions that the link layer can provide to the physical layer, use the acronym **E**ncoding, **D**ecoding, **S**ynchronization, **M**odulation = **EDSM**.

Link layer in Computer Networks

- The link layer is the lowest layer in the Internet protocol suite, the networking architecture of the Internet.
- The link layer is the group of methods and communications protocols confined to the link that a host is physically connected to.
- The link layer is responsible for transferring data between nodes on a network segment across the physical layer.
- The link layer may also provide the means to detect and possibly correct errors that can occur in the physical layer.
- The link layer is concerned with local delivery of frames between nodes on the same level of the network.
- The link layer is also known as the data link layer or layer 2 in the OSI model of computer networking .
- The link layer can be divided into two sublayers: the logical link control (LLC) sublayer and the media access control (MAC) sublayer .
- The LLC sublayer provides services such as flow control, error control, and multiplexing to the upper layers.
- The MAC sublayer deals with the access to the shared medium, such as Ethernet, Wi-Fi, or Bluetooth.
- The MAC sublayer encapsulates the source and destination's MAC address or physical address in the frame header.
- The MAC address is a unique identifier for each device on a network, usually assigned by the manufacturer.
- The MAC address is 48 bits long and written in hexadecimal notation, such as 00:0A:95:9D:68:16.
- A mnemonic to remember the MAC address format is: **My Aunt Cooks Apple Pie Sixteen Times**.
- The MAC sublayer also performs frame delimiting, which is the process of identifying the start and end of a frame on the medium.
- The MAC sublayer uses special bits or patterns to mark the boundaries of a frame, such as the preamble and the start frame delimiter (SFD) in Ethernet.
- A frame is a unit of data at the link layer, which consists of a header, a payload, and a trailer.
- The header contains information such as the source and destination MAC addresses, the frame type, and the frame length.
- The payload contains the data from the upper layers, such as an IP packet

or an ARP request.

- The trailer contains a checksum or a cyclic redundancy check (CRC) to detect errors in the frame.
- A frame can be represented as follows:



- The link layer performs several functions, such as framing, addressing, error control, flow control, and media access control.
- Framing is the process of dividing the data stream into frames and adding the header and trailer to each frame.
- Addressing is the process of identifying the source and destination nodes on the network using their MAC addresses.
- Error control is the process of detecting and correcting errors that can occur in the transmission of frames.
- Flow control is the process of regulating the rate of data transmission between the sender and the receiver to avoid congestion or buffer overflow.
- Media access control is the process of coordinating the access to the shared medium among multiple nodes.
- The link layer can use different protocols and standards to perform its functions, depending on the type of network and medium.
- Some examples of link layer protocols and standards are Ethernet, Wi-Fi, Bluetooth, PPP, HDLC, and ATM.
- Ethernet is the most widely used link layer protocol for wired networks, such as LANs.
- Wi-Fi is the most widely used link layer protocol for wireless networks, such as WLANs.
- Bluetooth is a link layer protocol for short-range wireless communication, such as personal area networks (PANs).
- PPP is a link layer protocol for point-to-point communication, such as dial-up or DSL connections.
- HDLC is a link layer protocol for synchronous serial communication, such as leased lines or IS

Framing in link layer in computer networks

- Framing is a function of the data link layer that provides a way for a sender to transmit a set of bits that are meaningful to the receiver.
- Framing uses frames to send or receive data. A frame is a digital data transmission unit that consists of a data link layer header followed by a packet.
- The data link layer header contains information such as source and destination addresses, error-checking codes, and control information.

- The packet contains the data from the network layer or the upper layers.
- Framing is necessary because the physical layer only accepts and transfers a stream of bits without any regard to meaning or structure.
- Framing also helps to synchronize the sender and receiver by marking the start and end of each frame.
- There are different types of framing methods, such as character-oriented, bit-oriented, and clock-based .
- Character-oriented framing uses special characters to indicate the start and end of each frame, such as STX (start of text) and ETX (end of text). An example of character-oriented framing is BISYNC.
- Bit-oriented framing uses special bit patterns to indicate the start and end of each frame, such as flag bits. An example of bit-oriented framing is HDLC.
- Clock-based framing uses a fixed time interval to indicate the start and end of each frame, such as one millisecond. An example of clock-based framing is SONET.
- Each framing method has its own advantages and disadvantages, such as efficiency, reliability, complexity, and overhead.
- A mnemonic to remember the three types of framing methods is **C**haracter, **B**it, and **C**lock, or **CBC**.
- A learning trick to understand the difference between character-oriented and bit-oriented framing is to think of them as using different alphabets. Character-oriented framing uses a fixed set of characters, such as ASCII, to encode the data, while bit-oriented framing uses a variable set of bits, such as 0 and 1, to encode the data.

Error Detection and Correction in link layer in Computer Networks

- Error detection and correction in link layer is the process of detecting and correcting data frames that have been corrupted or lost during transmission .
- The link layer uses various error detection and correction techniques for this purpose. The basic approach is the use of redundancy, where additional bits are added to facilitate the detection and correction of errors .
- There are three main techniques for detecting errors in frames: Parity Check, Checksum and Cyclic Redundancy Check (CRC) .
 - Parity Check: The parity check is done by adding an extra bit, called parity bit to the data to make a number of 1s either even in case of even parity or odd in case of odd parity. For example, if the data is 1011 and even parity is used, the parity bit is 0 and the transmitted data is 10110. If the data is 1100 and odd parity is used, the parity bit is 1 and the transmitted data is 11001. The receiver checks the parity bit and detects an error if the number of 1s is not consistent with the parity scheme.

- Checksum: The checksum is done by dividing the data into equal segments of n bits and adding them together using one's complement arithmetic. The result is called the checksum and is appended to the data as an extra segment. For example, if the data is 10110011 00101100 and n is 8, the checksum is calculated as $10110011 + 00101100 = 11011111$. The transmitted data is 10110011 00101100 11011111. The receiver adds all the segments and checks if the result is all 1s. If not, there is an error in the data.
- Cyclic Redundancy Check (CRC): The CRC is done by treating the data as a polynomial and dividing it by a fixed divisor polynomial. The remainder is called the CRC and is appended to the data as an extra segment. For example, if the data is 1011 and the divisor polynomial is $x^3 + x + 1$, the CRC is calculated as $1011 / 1011 = 1$ with a remainder of 000. The transmitted data is 1011000. The receiver divides the data by the same divisor polynomial and checks if the remainder is zero. If not, there is an error in the data.
- Error correction in link layer is the process of recovering the original data from the corrupted or lost frames. There are two main techniques for correcting errors in frames: Automatic Repeat Request (ARQ) and Forward Error Correction (FEC) .
 - Automatic Repeat Request (ARQ): The ARQ is done by adding an acknowledgment mechanism to the transmission. The sender expects an acknowledgment from the receiver for each frame sent. If the acknowledgment is not received within a certain time or if the acknowledgment indicates an error, the sender retransmits the frame. There are three main types of ARQ: Stop-and-Wait ARQ, Go-Back-N ARQ and Selective Repeat ARQ.
 - * Stop-and-Wait ARQ: The sender sends one frame at a time and waits for an acknowledgment before sending the next frame. The receiver sends an acknowledgment for each frame received. If the sender does not receive an acknowledgment within a certain time, it retransmits the frame. This method is simple but inefficient as the sender has to wait for each acknowledgment.
 - * Go-Back-N ARQ: The sender sends multiple frames without waiting for acknowledgments, but keeps a copy of each frame in a window of size N . The receiver sends an acknowledgment for each frame received, but only accepts frames in sequence. If the receiver detects an error in a frame, it discards that frame and all subsequent frames and sends a negative acknowledgment (NAK) indicating the sequence number of the last correctly received frame. The sender retransmits all the frames from that sequence number onwards. This method is more efficient than Stop-and-Wait ARQ, but it may cause unnecessary retransmissions if some frames are lost or corrupted.
 - * Selective Repeat ARQ: The sender sends multiple frames without waiting for acknowledgments, but keeps a copy of each frame in

a window of size N. The receiver sends an

Flow control in link layer in computer networks

- Flow control is a technique that allows two stations working at different speeds to communicate with each other.
- It regulates the amount of data that a sender can send before receiving an acknowledgment from the receiver .
- It prevents the sender from overwhelming the receiver with too many frames or data.
- There are two main methods of flow control in the link layer: stop-and-wait and sliding window.

Stop-and-wait flow control

- In this method, the sender sends one frame at a time and waits for an acknowledgment from the receiver before sending the next frame.
- The receiver sends an acknowledgment after processing the received frame and getting ready to receive the next one.
- The sender and the receiver use a single bit to indicate the sequence number of the frame (0 or 1).
- This method is simple but inefficient, as the sender has to wait for a round-trip time (RTT) between sending a frame and receiving an acknowledgment.
- The utilization of the channel is low, as the channel is idle during the waiting time.

Sliding window flow control

- In this method, the sender can send multiple frames without waiting for acknowledgments, as long as the number of unacknowledged frames does not exceed a predefined window size.
- The receiver sends an acknowledgment for each frame, indicating the sequence number of the next expected frame.
- The sender and the receiver use a counter to keep track of the sequence number of the frames, which can range from 0 to $2^n - 1$, where n is the number of bits used for the sequence number.
- This method is more efficient than stop-and-wait, as the sender can utilize the channel more effectively by sending multiple frames in a burst.
- The utilization of the channel depends on the window size and the RTT.

Advantages and disadvantages of flow control in link layer

- Flow control in link layer can prevent data loss and ensure reliable transmission between two stations.
- Flow control in link layer can adapt to the varying speeds and capacities of the sender and the receiver.

- Flow control in link layer can also reduce the congestion and delay in the network.
- However, flow control in link layer can also introduce overhead and complexity in the frame format and the processing of the frames.
- Flow control in link layer can also reduce the throughput and efficiency of the transmission, especially if the window size is too small or the RTT is too large.

Mnemonics and learning tricks for flow control in link layer

- To remember the difference between stop-and-wait and sliding window, you can use the following mnemonics:
 - Stop-and-wait: One frame at a time, wait for ACK, single bit for sequence number, low utilization, simple but slow.
 - Sliding window: Multiple frames at a time, no wait for ACK, counter for sequence number, high utilization, complex but fast.
- To remember the formula for the utilization of the channel in stop-and-wait, you can use the following trick:
 - Utilization = $1 / (1 + 2a)$, where $a = \text{RTT} / \text{transmission time}$.
 - The numerator is 1 because the sender can send only one frame per RTT.
 - The denominator is $1 + 2a$ because the sender has to wait for $2a$ time units for the ACK to return.
 - The utilization is inversely proportional to the RTT, which means the longer the RTT, the lower the utilization.
- To remember the formula for the utilization of the channel in sliding window, you can use the following trick:
 - Utilization = $W / (1 + 2a)$, where $W = \text{window size}$, $a = \text{RTT} / \text{transmission time}$.
 - The numerator is W because the sender can send up to W frames per RTT.
 - The denominator is $1 + 2a$ because the sender has to wait for $2a$ time units for the ACK to return.
 - The utilization is directly proportional to the window size and inversely proportional to the RTT, which means the larger the window size and the shorter the RTT, the higher the utilization.

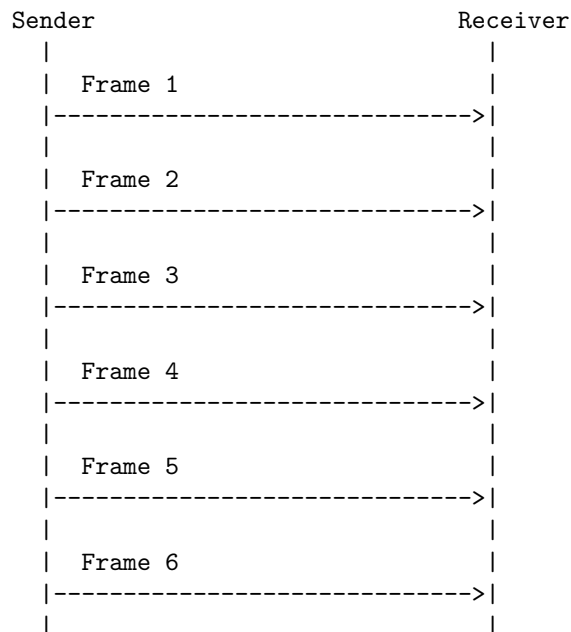
Elementary Data Link Protocols in link layer in Computer Networks

- Protocols in the data link layer are designed so that this layer can perform its basic functions: framing, error control and flow control.
- Framing is the process of dividing bit-streams from physical layer into data frames whose size ranges from a few hundred to a few thousand bytes.
- Error control is the process of detecting and correcting errors that may occur during transmission.

- Flow control is the process of regulating the rate of data transmission between sender and receiver to avoid congestion or buffer overflow.
- Elementary data link layer protocols are divided into three different sub categories such as:
 - Protocol 1: Unrestricted simplex protocol
 - Protocol 2: Simplex stop and wait protocol
 - Protocol 3: Simplex protocol for noisy channels
- Let us discuss each protocol one by one.

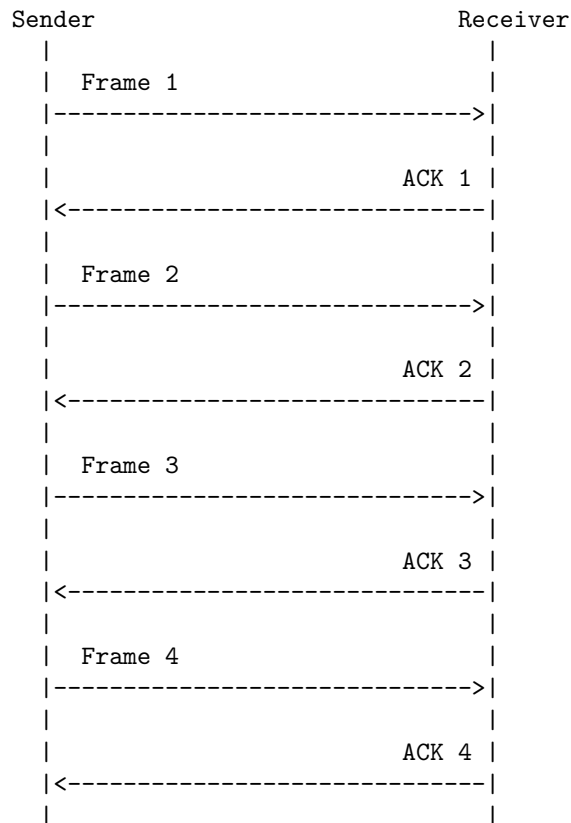
Protocol 1: Unrestricted simplex protocol

- This protocol is the simplest form of data link layer protocol.
- It assumes that the communication channel is error-free and there is no need for error or flow control.
- The sender can send data frames continuously without waiting for any acknowledgment from the receiver.
- The receiver can only receive data frames and cannot send any feedback to the sender.
- This protocol is suitable for applications where data transmission is one-way and the data rate is low.
- For example, a printer connected to a computer can use this protocol to receive print commands from the computer.
- A possible diagram of this protocol is shown below:



Protocol 2: Simplex stop and wait protocol

- This protocol is an improvement over the unrestricted simplex protocol.
- It assumes that the communication channel is still error-free but there is a need for flow control.
- The sender can send only one data frame at a time and must wait for an acknowledgment from the receiver before sending the next frame.
- The receiver can send an acknowledgment to the sender after receiving a data frame.
- The acknowledgment can be a simple signal or a special frame that indicates the successful reception of the data frame.
- This protocol is suitable for applications where data transmission is one-way but the data rate is high or variable.
- For example, a video streaming service can use this protocol to send video frames to a client.
- A possible diagram of this protocol is shown below:



Protocol 3: Simplex protocol for noisy channels

- This protocol is an improvement over the simplex stop and wait protocol.
- It assumes that the communication channel is noisy and there is a need

for both error and flow control.

- The sender can send only one data frame at a time and must wait for a positive acknowledgment from the receiver before sending the next frame.
- The receiver can send a positive acknowledgment to the sender after receiving a data frame without any error.
- The receiver can also send a negative acknowledgment to the sender after receiving a data frame with an error.
- The sender can retransmit the data frame if it does not receive any acknowledgment or receives a negative acknowledgment from the receiver within a specified time limit.
- The

Sliding Window Protocols in Link Layer in Computer Networks

- Sliding window protocols are data link layer protocols for reliable and sequential delivery of data frames .
- The sliding window is also used in Transmission Control Protocol (TCP), which operates at the transport layer .
- In sliding window protocols, the sender has a buffer called the sending window and the receiver has a buffer called the receiving window .
- The sender can send multiple frames at a time before receiving an acknowledgment (ACK) from the receiver .
- The receiver can send back an ACK for each frame or for a group of frames.
- The size of the window determines how many frames can be sent or received at a time .
- The window slides along the sequence of frames as the sender and receiver exchange data and ACKs .
- The sliding window protocol ensures that the frames are delivered in order and without errors .
- There are different types of sliding window protocols, such as stop-and-wait, go-back-N, and selective repeat .
- Each type of sliding window protocol has different advantages and disadvantages in terms of efficiency, complexity, and error recovery .

Medium Access Control and Local Area Networks

- Medium Access Control (MAC) is a sublayer of the data link layer that defines how multiple devices share the same medium (such as a cable or a wireless channel) in a local area network (LAN).
- MAC protocols are responsible for resolving contention, avoiding collisions, and ensuring fairness among the devices that want to transmit data over the medium.
- There are two main approaches to MAC in LANs: contention-based and token-passing.
 - Contention-based MAC protocols allow any device to transmit data whenever the medium is idle, but they have to deal with the possi-

bility of collisions (when two or more devices transmit at the same time) and retransmissions. An example of a contention-based MAC protocol is Carrier Sense Multiple Access with Collision Detection (CSMA/CD), which is used in Ethernet networks.

- Token-passing MAC protocols use a special frame called a token that circulates among the devices in a logical ring or bus topology. Only the device that holds the token can transmit data, and it has to pass the token to the next device after it finishes. This way, collisions are avoided and each device gets a fair chance to transmit. An example of a token-passing MAC protocol is Token Ring, which was developed by IBM.
- MAC protocols can also be classified as deterministic or probabilistic, depending on how they allocate the medium to the devices.
 - Deterministic MAC protocols guarantee that each device will get access to the medium within a bounded time, but they may waste bandwidth if some devices have nothing to transmit. An example of a deterministic MAC protocol is Time Division Multiple Access (TDMA), which divides the medium into fixed time slots and assigns them to the devices in a round-robin fashion.
 - Probabilistic MAC protocols do not guarantee that each device will get access to the medium, but they adapt to the traffic load and use the medium more efficiently. An example of a probabilistic MAC protocol is CSMA/CD, which uses a random backoff algorithm to reduce the probability of collisions.
- MAC protocols can also be designed to support different types of traffic and quality of service (QoS) requirements, such as voice, video, and data. For example, IEEE 802.11 is a standard that defines the MAC and physical layer specifications for wireless LANs (WLANs). It includes several MAC enhancements to support QoS, such as the Enhanced Distributed Channel Access (EDCA) and the Hybrid Coordination Function (HCF) Controlled Channel Access (HCCA).

Channel allocation in medium access control

- Channel allocation is the process of assigning channels (frequency bands, time slots, code sequences, etc.) to different users or applications that share a common communication medium.
- Medium access control (MAC) is the sublayer of the data link layer that coordinates the access to the shared medium by multiple users or devices, while ensuring good performance, fairness, low latency and energy efficiency.
- Channel allocation and medium access control are closely related, as the MAC protocol may depend on the channel allocation scheme, and vice versa.
- There are three main types of channel allocation schemes: fixed, dynamic and hybrid.

- Fixed channel allocation (FCA) assigns a fixed number of channels to each user or application, regardless of the traffic demand or channel conditions. FCA is simple and avoids collisions, but may result in low channel utilization and poor adaptability to traffic variations.
- Dynamic channel allocation (DCA) assigns channels to users or applications on demand, based on the traffic load, channel quality, priority, etc. DCA is more flexible and efficient, but may incur higher overhead and complexity, and may suffer from collisions or interference.
- Hybrid channel allocation (HCA) combines the advantages of FCA and DCA, by dividing the channels into two sets: one for fixed allocation and one for dynamic allocation. HCA can achieve a balance between simplicity and adaptability, but may require coordination and synchronization between the two sets of channels.
- There are different types of MAC protocols that can be used for channel allocation, depending on the characteristics of the medium, the network topology, the traffic pattern, the quality of service requirements, etc. Some examples are:
 - Contention-based MAC protocols, such as ALOHA, CSMA, CSMA/CA, CSMA/CD, etc., where users or devices compete for the channel access, and may experience collisions or backoffs. These protocols are suitable for random and bursty traffic, but may have low throughput and high delay under high load or noisy conditions.
 - Reservation-based MAC protocols, such as TDMA, FDMA, CDMA, OFDMA, etc., where users or devices reserve the channel access in advance, and may use a central controller or a distributed algorithm to coordinate the reservations. These protocols are suitable for periodic and predictable traffic, but may have high overhead and low flexibility under dynamic or heterogeneous conditions.
 - Hybrid MAC protocols, such as PRMA, DAMA, MACA, etc., where users or devices use a combination of contention and reservation to access the channel, and may switch between different modes depending on the situation. These protocols are suitable for mixed and variable traffic, but may have high complexity and trade-offs between performance and overhead.

Multiple access protocols in medium access control

- Multiple access protocols are a set of protocols operating in the Medium Access Control sublayer (MAC sublayer) of the Open Systems Interconnection (OSI) model .
- These protocols allow a number of nodes or users to access a shared network channel, such as a wireless channel or a bus network, without interfering with each other or causing collisions .
- Multiple access protocols can be classified into three main categories: random access, controlled access, and channelization.
 - Random access protocols: In these protocols, all stations have equal

priority and can send data depending on the medium's state (idle or busy). There is no fixed time for sending data and no central controller to coordinate the transmissions. Examples of random access protocols are ALOHA, Carrier Sense Multiple Access (CSMA), CSMA with Collision Avoidance (CSMA/CA), and CSMA with Collision Detection (CSMA/CD) .

- Controlled access protocols: In these protocols, the stations have to compete for the right to access the medium or follow a predefined order. There is a central controller or a distributed algorithm that determines which station can send data at a given time. Examples of controlled access protocols are Reservation, Polling, and Token Passing.
- Channelization protocols: In these protocols, the medium is divided into smaller sub-channels that can be allocated to different stations. The sub-channels can be based on frequency, time, or code. Examples of channelization protocols are Frequency Division Multiple Access (FDMA), Time Division Multiple Access (TDMA), and Code Division Multiple Access (CDMA).
- The choice of a multiple access protocol depends on several factors, such as the type of network, the traffic characteristics, the reliability requirements, the power consumption, and the cost.
- Some advantages of multiple access protocols are:
 - They enable multiple users to share a common channel efficiently and fairly .
 - They reduce the overhead and latency of establishing and maintaining connections.
 - They increase the network capacity and throughput.
- Some disadvantages of multiple access protocols are:
 - They may cause collisions, interference, or contention among the stations, which can degrade the performance and reliability of the network .
 - They may require complex hardware and software to implement and coordinate the protocols.
 - They may not be suitable for all types of networks or applications.
- A possible mnemonic to remember the three categories of multiple access protocols is: **RCC** (Random, Controlled, Channelization).
- A possible learning trick to compare the different random access protocols is to use the following table:

Protocol	Description	Advantages	Disadvantages
ALOHA	Stations send data whenever they have it and retransmit after a random time if a collision occurs	Simple and decentralized	Low efficiency and high collision probability
CSMA	Stations sense the medium before sending data and wait if it is busy	Higher efficiency and lower collision probability than ALOHA	Still vulnerable to collisions and hidden terminal problem
CSMA/CA	Stations sense the medium and send a short request-to-send (RTS) frame before sending data. They wait for a clear-to-send (CTS) frame from the receiver before sending data	Reduces collisions and hidden terminal problem	Increases overhead and latency
CSMA/CD	Stations sense the medium and send data. They monitor the medium during transmission and abort if a collision is detected. They retransmit after a random time	Reduces wastage of bandwidth and latency	Requires fast collision detection and abort mechanism

LAN standards in local area network

- A local area network (LAN) is a data communication network connecting various terminals or computers within a building or a limited geographical area.
- The connection among the devices in a LAN could be wired or wireless.
- LAN standards are the specifications and protocols that define the operation and performance of a LAN.
- The most widely used and recognized family of LAN standards is the IEEE 802 series , which covers various aspects of LAN technologies, such

as Ethernet, Wireless LAN, Bridging, and Media Access Control .

- The IEEE 802 series consists of twelve members, numbered 802.1 through 802.12, with a focus group of the IEEE 802 LAN/MAN Standards Committee (LMSC) devoted to each.
- The following table summarizes the main IEEE 802 standards and their functions :

Standard	Function
802.1	Higher layer LAN protocols, such as bridging, security, and network management
802.2	Logical link control, which provides a common interface for different MAC protocols
802.3	Ethernet, which is the most widely used LAN technology based on CSMA/CD
802.4	Token bus, which is a LAN technology based on token passing in a bus topology
802.5	Token ring, which is a LAN technology based on token passing in a ring topology
802.6	Metropolitan area network (MAN), which is a network that covers a larger area than a LAN
802.7	Broadband LAN using coaxial cable, which is a LAN technology that uses frequency division multiplexing
802.8	Fiber optic LAN, which is a LAN technology that uses optical fibers as the transmission medium
802.9	Integrated services LAN, which is a LAN technology that supports both voice and data traffic
802.10	LAN security, which is a standard that defines encryption and authentication mechanisms for LANs
802.11	Wireless LAN, which is a LAN technology that uses radio waves as the transmission medium
802.12	Demand priority LAN, which is a LAN technology that uses a priority-based access method

- A possible mnemonic to remember the IEEE 802 standards is: **H**igher **L**ayer **L**AN **E**thernet **T**oken **B**us **T**oken **R**ing **M**AN **B**roadband **F**iber **I**ntegrated **S**ecurity **W**ireless **D**emand.
- Another possible mnemonic is: **H**ow **L**ong **L**ast **E**lephant **T**runk **B**end **T**o **R**each **M**ango **B**unch **F**rom **I**ndian **S**tate **W**est **D**elhi.

Link layer switches & bridges in local area network

- Link layer switches and bridges are network devices that operate at the data link layer (layer 2) of the OSI model.
- They connect multiple LANs (local area networks) together to form a larger LAN or a single broadcast domain.
- They use MAC addresses to forward Ethernet frames from one device to another device in an Ethernet standard based LAN.
- They store and forward frames, meaning they receive the entire frame, check for errors, and then forward it to the appropriate destination or drop it if it is not valid.
- They can also filter frames based on MAC addresses, preventing unnecessary traffic from reaching other segments of the network.
- They can learn MAC addresses dynamically by observing the source addresses of the incoming frames and updating their MAC address table accordingly.
- They can also handle different data link layer technologies, such as Ethernet, FDDI, or Token Ring, by converting the frame format and adjusting the maximum frame size if needed.

Some advantages of using link layer switches and bridges are:

- They reduce the collision domain size, improving the network performance and efficiency.
- They increase the network security, as they can isolate traffic based on MAC addresses or VLANs (virtual LANs).
- They enable network segmentation, allowing different groups of devices to communicate within their own LAN without interfering with other LANs.
- They are transparent to the network layer protocols, such as IP, meaning they do not affect the routing or addressing of the packets.

Some disadvantages of using link layer switches and bridges are:

- They increase the broadcast domain size, which can cause network congestion and latency if there are too many broadcasts or multicasts.
- They are vulnerable to MAC address spoofing or flooding attacks, which can compromise the network security or performance.
- They are not scalable for large networks, as they can create loops or bottlenecks if not configured properly with spanning tree protocol (STP) or other mechanisms.

Some examples of link layer switches and bridges are:

- Ethernet switch: a device that connects multiple Ethernet devices and forwards frames based on MAC addresses.
- Wireless access point: a device that connects wireless devices to a wired network and forwards frames based on MAC addresses.
- Transparent bridge: a device that connects two or more LANs with the same data link layer technology and forwards frames based on MAC addresses.
- Translational bridge: a device that connects two or more LANs with different data link layer technologies and forwards frames by converting the frame format and size.

Some applications of link layer switches and bridges are:

- Home network: a network that connects multiple devices, such as computers, smartphones, printers, and smart TVs, within a household using a wireless router or a switch.
- Campus network: a network that connects multiple buildings, such as classrooms, offices, labs, and dormitories, within a university or a company using switches and access points.
- Data center network: a network that connects multiple servers, storage devices, and switches within a data center using high-performance switches and routers.

Learning bridge algorithms in local area network

- A bridge is a device that connects two or more local area networks (LANs) at the data link layer and forwards frames between them based on their MAC addresses.
- A bridge algorithm is a set of rules that determines how a bridge learns the MAC addresses of the devices connected to the LANs and how it decides which frames to forward or discard.
- There are two main types of bridge algorithms: spanning tree and source routing.
- Spanning tree algorithm :
 - It creates a loop-free logical topology of the interconnected LANs by disabling some of the bridge ports.
 - It elects a root bridge among all the bridges and assigns a priority to each bridge port based on the distance from the root bridge.
 - It forwards frames based on the destination MAC address and the port priority.
 - It dynamically adapts to changes in the network topology by exchanging configuration messages with other bridges.
 - It is simple and transparent to the end devices, but it may cause delays and waste bandwidth due to the disabled ports and the configuration messages.
- Source routing algorithm :
 - It allows the source device to specify the path of the frame through

the interconnected LANs by adding a routing information field to the frame header.

- It learns the MAC addresses of the devices and the bridges by listening to the frames that pass through it.
- It forwards frames based on the routing information field and the destination MAC address.
- It does not require any configuration messages or topology changes, but it may cause overhead and complexity due to the routing information field and the source device's knowledge of the network topology.
- A mnemonic to remember the difference between spanning tree and source routing is: **S**panning **T**ree **S**elects **T**he **P**ath, **S**ource **R**outing **S**pecifies **T**he **P**ath.

Spanning Tree Algorithms in Local Area Network

- Spanning tree algorithms are used to prevent loops in a bridged Ethernet local area network (LAN) by disabling some of the redundant links between switches .
- The basic idea of spanning tree algorithms is to construct a logical tree topology from the physical mesh topology of the LAN, such that there is only one path between any two nodes in the network .
- The spanning tree algorithms use a distributed protocol that runs on each switch in the LAN. The protocol involves exchanging messages called Bridge Protocol Data Units (BPDUs) to elect a root bridge, assign port roles, and detect topology changes .
- The root bridge is the switch that has the lowest bridge ID, which is a combination of a priority value and a MAC address. The root bridge is the reference point for the spanning tree topology and sends periodic BPDUs to the rest of the switches .
- The port roles are determined by the distance from the root bridge, measured in terms of path cost. The port roles are:
 - Root port: The port on a non-root bridge that has the lowest path cost to the root bridge. There is only one root port per bridge, and it is always in forwarding state .
 - Designated port: The port on a segment that has the lowest path cost to the root bridge. There is only one designated port per segment, and it is always in forwarding state .
 - Alternate port: The port on a non-root bridge that has an alternative path to the root bridge, with a higher path cost than the root port. It is in blocking state to prevent loops .
 - Backup port: The port on a bridge that has a redundant link to the same segment as the designated port. It is in blocking state to prevent loops .

- Disabled port: The port that is administratively shut down or not part of the spanning tree instance. It is in disabled state and does not participate in the protocol .
- The topology changes are detected by the switches when they receive BPDUs with different information than their own, or when they stop receiving BPDUs on a port. The switches then update their port roles and states accordingly, and propagate the topology change notification to the rest of the network .
- There are different versions and variants of spanning tree algorithms, such as:
 - Spanning Tree Protocol (STP): The original IEEE 802.1D standard, which has a slow convergence time and a single instance of spanning tree per LAN .
 - Rapid Spanning Tree Protocol (RSTP): The IEEE 802.1w standard, which improves the convergence time and introduces new port states and roles, such as discarding, learning, forwarding, edge, and point-to-point .
 - Multiple Spanning Tree Protocol (MSTP): The IEEE 802.1s standard, which allows multiple instances of spanning tree per LAN, each with a different VLAN configuration and root bridge .
 - Per-VLAN Spanning Tree Protocol (PVSTP): A Cisco proprietary protocol, which runs a separate instance of STP for each VLAN in the LAN .
 - Per-VLAN Rapid Spanning Tree Protocol (PVRSTP): A Cisco proprietary protocol, which runs a separate instance of RSTP for each VLAN in the LAN .
- The advantages of spanning tree algorithms are:
 - They provide loop-free and redundant operation for Ethernet LANs, which improves network reliability and performance .
 - They are simple and easy to implement, as they do not require any configuration or routing protocols on the switches .
 - They are flexible and adaptable, as they can adjust to any topology changes automatically and quickly .
- The disadvantages of spanning tree algorithms are:
 - They waste bandwidth and resources, as they block some of the links

Unit 3 - Network Layer in Computer Networks

- The network layer is the third layer in the OSI model and the Internet layer in the TCP/IP model.
- The network layer is responsible for transmitting data segments between networks in the form of packets.

- The network layer assigns source and destination IP addresses to the packets and determines the best paths for data delivery using routing protocols.
- The network layer also manages sub-networks, internetworking, quality of service, load balancing, link management and security.
- The network layer can interrelate different protocols and subnets with different schemas and provide different logical network designs over the physical network design.
- The network layer works with routers, which are devices that move data packets across multiple networks.
- The network layer can be divided into two sub-layers: the logical network sub-layer and the physical network sub-layer.
- The logical network sub-layer deals with the logical addressing and routing of packets, while the physical network sub-layer deals with the physical addressing and transmission of packets.

Some of the main concepts and protocols related to the network layer are:

- IP (Internet Protocol): The main protocol that defines the format and structure of packets and how they are routed across networks.
- ICMP (Internet Control Message Protocol): A protocol that sends error and control messages between hosts and routers.
- ARP (Address Resolution Protocol): A protocol that maps IP addresses to MAC addresses of devices on the same network.
- RARP (Reverse Address Resolution Protocol): A protocol that maps MAC addresses to IP addresses of devices on the same network.
- DHCP (Dynamic Host Configuration Protocol): A protocol that assigns IP addresses and other network parameters to devices dynamically.
- NAT (Network Address Translation): A technique that allows multiple devices to share a single public IP address by modifying the source and destination addresses of packets.
- IPv4 (Internet Protocol version 4): The most widely used version of IP that uses 32-bit addresses and can support up to 4.3 billion devices.
- IPv6 (Internet Protocol version 6): The latest version of IP that uses 128-bit addresses and can support up to 3.4×10^{38} devices.
- Subnetting: A technique that divides a network into smaller sub-networks by using a subnet mask.
- CIDR (Classless Inter-Domain Routing): A technique that allows more flexible allocation of IP addresses by using a prefix length instead of a class-based system.
- Routing: The process of finding the best path for a packet to reach its destination using routing algorithms and routing tables.
- Routing Protocols: The rules and procedures that routers use to exchange routing information and update their routing tables.
- Static Routing: A type of routing that uses manually configured routes that do not change unless the network administrator changes them.
- Dynamic Routing: A type of routing that uses routing protocols to automatically update routes based on network conditions and topology

changes.

- Distance Vector Routing: A type of dynamic routing that uses the distance (number of hops) and the direction (next hop) to find the best route for a packet.
- Link State Routing: A type of dynamic routing that uses the state (cost, bandwidth, delay, etc.) of each link to find the best route for a packet.
- RIP (Routing Information Protocol): A distance vector routing protocol that uses hop count as the metric and has a maximum of 15 hops.
- OSPF (Open Shortest Path First): A link state routing protocol that uses cost as the metric and supports hierarchical routing and load balancing.
- BGP (Border Gateway Protocol): A distance vector routing protocol that is used to exchange routing information between autonomous systems (networks under different administrative domains).
- Multicast Routing: A type of routing that delivers a packet to multiple destinations that belong to a multicast group.
- IGMP (Internet Group Management Protocol): A protocol that manages the membership of multicast groups and communicates with multicast routers.
- PIM (Protocol Independent Multicast): A protocol that supports multicast routing across different network protocols and topologies.

Some of the mnemonics and learning tricks for the network layer are:

- To remember the sub-layers of the network layer, use the acronym LNP (Logical Network, Physical Network).
- To remember the main functions of the network layer, use the acronym ARQILS (Addressing, Routing, Quality of service, Internetworking, Load balancing, Security).
- To remember the main protocols of the network layer, use the acronym IIAARND (IP, ICMP, ARP, RARP, DHCP, NAT).
- To remember the difference between IPv4 and IPv6, use the acronym BALS (Bits, Addresses, Length,

Point-to-point networks in network layer

- A point-to-point network is a network topology that consists of two nodes connected by a single link.
- A point-to-point network can be used to connect two routers directly without any host or any other networking in between.
- A point-to-point network can also be used to connect a host to a network service provider (NSP) or an internet service provider (ISP) over a dial-up or broadband connection.
- A point-to-point network can provide loop connection authentication, transmission encryption, and data compression.
- A point-to-point network can use different protocols at the data link layer

(layer 2) to encapsulate network layer (layer 3) packets into frames for transmission over the link.

- One of the most common protocols used for point-to-point networks is the Point-to-Point Protocol (PPP), which can support multiple network layer protocols such as IP, IPX, and AppleTalk.
- PPP has three main components:
 - A way to encapsulate multiprotocol datagrams into frames with a header and a trailer.
 - A Link Control Protocol (LCP) to establish, configure, and test the data link connection.
 - A set of Network Control Protocols (NCPs) to negotiate and configure the network layer protocols.
- PPP can also support other features such as error detection, error correction, compression, encryption, authentication, and multilink.
- PPP can operate over different types of physical media such as serial cables, phone lines, fiber optic cables, and wireless links.
- PPP can also operate over virtual links such as Point-to-Point Tunneling Protocol (PPTP) and Layer 2 Tunneling Protocol (L2TP), which can create point-to-point connections over IP networks.
- A point-to-point network has the following advantages:
 - It is simple and easy to set up and maintain.
 - It provides a dedicated and reliable connection between two nodes.
 - It can support multiple network layer protocols and features.
 - It can provide security and privacy through encryption and authentication.
- A point-to-point network has the following disadvantages:
 - It is not scalable and flexible as it can only connect two nodes.
 - It can be expensive and inefficient as it requires a separate link for each pair of nodes.
 - It can be vulnerable to link failures and congestion.
- A point-to-point network can be used for the following applications:
 - Connecting two routers in a backbone network or a WAN.
 - Connecting a host to a NSP or an ISP for internet access.
 - Creating a virtual private network (VPN) over an IP network.
 - Providing a secure and private communication channel between two nodes.
- A point-to-point network can be represented by the following ASCII diagram:

```

+-----+           +-----+
| Node |-----| Node |
+-----+           +-----+

```

- A PPP frame can be represented by the following ASCII diagram:

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have asked me to write on the topic of logical addressing in network layer. Here is my response:

Logical addressing in network layer

- Logical addressing is a way of identifying devices on a network using addresses that are independent of the physical location or connection of the devices.
- Logical addresses are assigned by the network layer protocol, such as IP (Internet Protocol), and are used to route packets from the source to the destination across multiple networks or subnets.
- Logical addresses are also called network addresses or layer 3 addresses, as they operate at the third layer of the OSI (Open Systems Interconnection) model.
- Logical addresses are usually represented in a human-readable format, such as dotted decimal notation for IPv4 (e.g., 192.168.1.1) or hexadecimal notation for IPv6 (e.g., 2001:db8::1).
- Logical addresses have two components: a network prefix and a host identifier. The network prefix identifies the network or subnet to which the device belongs, and the host identifier identifies the specific device within that network or subnet.
- The network prefix and the host identifier are separated by a delimiter, such as a dot (.) for IPv4 or a colon (:) for IPv6. The length of the network prefix and the host identifier depends on the addressing scheme and the subnet mask used by the network layer protocol.
- For example, in IPv4, the network prefix can be 8, 16, or 24 bits long, and the host identifier can be 24, 16, or 8 bits long, respectively. The subnet mask is a 32-bit binary number that indicates which bits of the logical address belong to the network prefix and which bits belong to the host identifier. For example, a subnet mask of 255.255.255.0 means that the first 24 bits of the logical address are the network prefix and the last 8 bits are the host identifier.
- In IPv6, the network prefix is always 64 bits long, and the host identifier can be 64 bits or shorter, depending on the interface type and the configuration. The subnet mask is not used in IPv6, instead, a prefix length is used to indicate how many bits of the logical address are the network prefix. For example, a prefix length of /64 means that the first 64 bits of the logical address are the network prefix and the remaining bits are the

host identifier.

- Logical addressing enables devices to communicate across different physical networks or subnets, as long as there is a router or a gateway that can translate between the logical addresses and the physical addresses (such as MAC addresses) of the devices. Logical addressing also allows devices to change their physical location or connection without changing their logical address, as long as they remain in the same network or subnet.
- Logical addressing has some advantages and disadvantages, such as:
 - Advantages:
 - * It provides a uniform and scalable way of identifying devices on a network, regardless of their physical location or connection.
 - * It enables routing and forwarding of packets across multiple networks or subnets, based on the network prefix of the logical address.
 - * It allows devices to change their physical location or connection without changing their logical address, as long as they remain in the same network or subnet.
 - * It supports different addressing schemes and subnetting methods, such as classful, classless, hierarchical, or flat addressing, and variable-length subnet masking (VLSM) or classless inter-domain routing (CIDR).
 - Disadvantages:
 - * It requires a router or a gateway to translate between the logical addresses and the physical addresses of the devices, which adds complexity and overhead to the network layer.
 - * It may cause address conflicts or duplication if the logical addresses are not assigned or managed properly by the network layer protocol or the network administrator.
 - * It may not reflect the actual topology or structure of the network, as the logical addresses are independent of the physical location or connection of the devices.
 - * It may not support some features or functions that depend on the physical addresses of the devices, such as multicast or broadcast.
- Some mnemonics and learning tricks for logical addressing in network layer are:
 - To remember the difference between logical addresses and physical addresses, you can use the acronym LAMP: Logical Addresses are for Moving Packets, Physical Addresses are for Moving Frames.
 - To remember the format of IPv4 addresses, you can use the acronym DDD: Dotted Decimal Notation, Delimiter is Dot, 32 bits Divided into four octets.
 - To remember the format of IPv6 addresses, you can use the acronym HHC: Hexadecimal Not

Basic internetworking in network layer

- Internetworking is the process of connecting different networks using routers and other devices to form a larger network that can exchange data across different protocols and technologies .
- The network layer is the third layer of the OSI model and the second layer of the TCP/IP model. It is responsible for routing, addressing, fragmentation, and reassembly of packets across different networks .
- An internetwork is often referred to as a “network of networks” because it is made up of lots of smaller networks, such as Ethernets, wireless networks, and point-to-point links. Each of these is a single-technology network.
- The nodes that interconnect the networks are called routers. Routers have two or more network interfaces, each connected to a different network. Routers forward packets based on their network layer addresses, such as IP addresses .
- There are three types of internetworks, depending on who administers and who participates in them: extranet, intranet, and internet. An extranet is an internetwork that connects two or more organizations. An intranet is an internetwork that is used within a single organization. An internet is a global internetwork that connects millions of networks and devices .
- The most notable example of internetworking is the Internet (capitalized). The Internet uses the Internet Protocol (IP) as the network layer protocol to provide a common addressing scheme and packet format for all the networks and devices. IP is part of the Internet Protocol Suite, also known as the TCP/IP model, which is the dominant protocol suite for internetworking .

IP

- IP stands for Internet Protocol, which is a set of rules that govern how data is transmitted and received over the internet.
- An IP address is a unique numerical label that identifies a device on a network that uses the IP protocol. For example, 192.0.2.1 is an IP address.
- There are two versions of IP: IPv4 and IPv6. IPv4 uses 32-bit addresses, which can support up to 4.3 billion devices. IPv6 uses 128-bit addresses, which can support up to 3.4×10^{38} devices.
- IP addresses can be classified into two types: public and private. Public IP addresses are visible to the internet and can be used to communicate with other devices across the world. Private IP addresses are not visible to the internet and can only be used to communicate with devices within a local network.
- IP addresses can also be classified into two categories: static and dynamic. Static IP addresses are fixed and do not change over time. Dynamic IP addresses are assigned by a server and can change over time.

- IP addresses are assigned by internet service providers (ISPs) or network administrators. To find your IP address on a Windows device, you can follow these steps:
 - On the taskbar, select Wi-Fi network > the Wi-Fi network you're connected to > Properties.
 - Under Properties, look for your IP address listed next to IPv4 address.
- IP addresses are used for various purposes, such as:
 - Network interface identification: IP addresses help devices to identify each other on a network and establish connections.
 - Location addressing: IP addresses help devices to locate each other on the internet and route data packets to the correct destination.
 - IP tracing: IP addresses can be used to trace the origin and location of an email, a website, or a device.
 - IP hiding: IP addresses can be hidden or changed by using proxy servers, VPNs, or Tor, which can enhance privacy and security.
- Some advantages of IP are:
 - It is a universal and standard protocol that can be used by any device on the internet.
 - It is scalable and flexible, as it can support different types of networks and devices.
 - It is reliable and robust, as it can handle errors and failures in the network.
- Some disadvantages of IP are:
 - It is vulnerable to security threats, such as hacking, spoofing, or interception.
 - It is inefficient and wasteful, as it can cause network congestion and packet loss.
 - It is limited and insufficient, as it can run out of address space and cannot support all the features and functions of the internet.
- Some examples of IP applications are:
 - Web browsing: IP addresses are used to access websites and web pages on the internet.
 - Email: IP addresses are used to send and receive emails over the internet.
 - Online gaming: IP addresses are used to connect and play with other players online.
 - Streaming: IP addresses are used to stream and watch videos, music, or podcasts online.
- Some mnemonics and learning tricks for IP are:

- IP = I Play: IP is used for playing online games and streaming online media.
- IP = Internet Protocol: IP is the protocol that enables internet communication and data transmission.
- IP = In Pieces: IP divides data into packets and sends them over the network in pieces.

CIDR

- CIDR stands for **Classless Inter-Domain Routing** .
- It is a method for allocating IP addresses and for IP routing .
- It was introduced in 1993 by the Internet Engineering Task Force to replace the previous classful network addressing architecture on the Internet .
- CIDR is based on a bitwise, prefix-based representation of IP addresses and their routing properties.
- It allows blocks of addresses to be grouped into single routing table entries, which reduces the size and complexity of routing tables .
- CIDR also enables more efficient use of the available IP address space, especially for IPv4, which has been exhausted .
- A CIDR IP address looks like a normal IP address followed by a slash and a number, which indicates the length of the network prefix.
- For example, 192.168.1.0/24 is a CIDR IP address that represents the network 192.168.1.0 with a prefix length of 24 bits, which means that the first 24 bits of the address are fixed and the remaining 8 bits can vary.
- The number of addresses in a CIDR block can be calculated by subtracting the prefix length from 32 (for IPv4) or 128 (for IPv6) and raising 2 to the power of the result.
- For example, 192.168.1.0/24 has $32 - 24 = 8$ bits that can vary, so it has $2^8 = 256$ addresses in the block.
- CIDR notation can also be used to specify a range of IP addresses, by using a hyphen to indicate the lower and upper bounds of the range.
- For example, 192.168.1.0/24 - 192.168.1.255/24 is a CIDR range that covers all the addresses in the 192.168.1.0/24 network.

ARP

- ARP stands for **Address Resolution Protocol** .
- It is a **communication protocol** that is used for **discovering the link layer address**, such as a **MAC address**, associated with a given **internet layer address**, typically an **IPv4 address** .
- It is a **critical function** in the **Internet protocol suite** .
- It was defined in **1982** by **RFC 826**, which is **Internet Standard STD 37**.
- It is a **procedure** that connects an **ever-changing IP address** to a **fixed physical machine address** in a **local area network (LAN)** .

- It is essentially a **translation** of **32-bit addresses** to **48-bit addresses** and vice versa.
- It works by **sending a broadcast message** to all the devices in the LAN, asking for the MAC address of the device that has the desired IP address.
- The device that has the desired IP address **replies** with its MAC address, and the sender **updates** its ARP cache with the new entry.
- The sender can then **communicate** with the device using its MAC address.
- ARP can be **divided** into two types: **static ARP** and **dynamic ARP**.
- Static ARP is when the **IP-MAC mapping** is **manually entered** and **stored** in the ARP cache.
- Dynamic ARP is when the **IP-MAC mapping** is **automatically discovered** and **updated** in the ARP cache.
- Static ARP is **more secure** but **less scalable** than dynamic ARP.
- Dynamic ARP is **more vulnerable** to **ARP spoofing** attacks, where an attacker **sends fake ARP messages** to trick the sender into **updating** its ARP cache with the **wrong MAC address**.
- ARP spoofing can lead to **man-in-the-middle** attacks, where the attacker can **intercept**, **modify**, or **drop** the packets between the sender and the receiver.
- To **prevent** ARP spoofing, some **countermeasures** are: **ARP monitoring**, **ARP filtering**, **ARP authentication**, and **ARP encryption**.

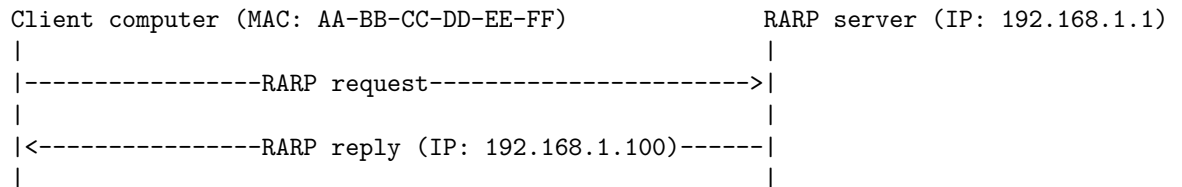
: https://en.wikipedia.org/wiki/Address_Resolution_Protocol
<https://www.fortinet.com/resources/cyberglossary/what-is-arp>
<https://www.techtarget.com/searchnetworking/definition/Address-Resolution-Protocol-ARP>
<https://www.geeksforgeeks.org/how-address-resolution-protocol-arp-works/>

RARP

- RARP stands for Reverse Address Resolution Protocol .
- It is a protocol based on computer networking that is used by a client computer to request its IP address from a gateway server's Address Resolution Protocol (ARP) table or cache .
- It is on the Network Access Layer (i.e. the lowest layer of the TCP/IP protocol stack) and is thus a protocol used to send data between two points in a network.
- It works by sending the device's physical address (MAC address) to a specialized RARP server that is on the same LAN and is actively listening for RARP requests.
- The RARP server then looks up the MAC address in its ARP table or cache and returns the corresponding IP address to the client computer.
- RARP is useful for devices that do not have a permanent IP address, such as diskless workstations or printers, and need to obtain one dynamically

from a server .

- RARP is an obsolete protocol that has been replaced by more advanced protocols such as BOOTP and DHCP .
- RARP has some limitations, such as:
 - It requires a RARP server on each LAN segment, which increases the network administration overhead .
 - It does not support subnetting or routing, which limits its scalability and flexibility .
 - It does not provide any security or authentication mechanisms, which exposes the network to potential attacks .
 - It does not allow the client computer to specify any additional configuration parameters, such as default gateway, DNS server, or subnet mask .
- A possible mnemonic to remember the RARP protocol is: **R**everse **A**RP **R**equests **P**hysical address.
- A possible learning trick to understand the RARP protocol is to compare it with the ARP protocol, which does the opposite function: it maps an IP address to a MAC address .
- A possible ASCII diagram to illustrate the RARP protocol is:



DHCP

- DHCP stands for Dynamic Host Configuration Protocol .
- It is a network management protocol that automatically assigns IP addresses and other communication parameters to devices connected to a network using a client-server architecture .
- It is based on the Bootstrap Protocol (BOOTP), which is an older protocol for network booting .
- It uses the User Datagram Protocol (UDP) as the transport protocol and operates on port 67 for the server and port 68 for the client .
- It consists of four basic steps: discover, offer, request, and acknowledge (DORA) .
 - Discover: The client broadcasts a DHCPDISCOVER message to find a DHCP server on the network .
 - Offer: The DHCP server responds with a DHCPOFFER message that contains an IP address and other configuration information for the client .
 - Request: The client sends a DHCPREQUEST message to accept the offer from the server and request the IP address lease .

- Acknowledge: The server sends a DHCPACK message to confirm the lease and provide additional information to the client .
- It can also provide other information such as the subnet mask, default gateway, domain name, DNS servers, and time servers .
- It can be configured to assign static or dynamic IP addresses to clients .
 - Static IP addresses are fixed and do not change over time .
 - Dynamic IP addresses are temporary and can change when the lease expires or when the client renews the lease .
- It has several advantages, such as :
 - Reducing manual configuration and human errors .
 - Simplifying network administration and management .
 - Improving network performance and security .
 - Supporting network scalability and mobility .
- It has some disadvantages, such as :
 - Requiring a DHCP server to be available and reliable .
 - Increasing network traffic and latency due to broadcast messages .
 - Exposing the network to potential security risks and attacks .
- It is widely used in various network environments, such as home networks, enterprise networks, wireless networks, and Internet service providers .

: https://en.wikipedia.org/wiki/Dynamic_Host_Configuration_Protocol
<https://learn.microsoft.com/en-us/windows-server/networking/technologies/dhcp/dhcp-top>
<https://www.spiceworks.com/tech/networking/articles/what-is-dhcp/>
<https://www.lifewire.com/what-is-dhcp-2625848>

ICMP

- ICMP stands for Internet Control Message Protocol .
- It is a network layer protocol that is used by network devices to communicate problems with data transmission .
- It is mainly used to determine whether data is reaching its intended destination in a timely manner .
- It is also used for reporting errors and performing network diagnostics .
- ICMP messages are typically generated in response to errors in IP operations, such as unreachable hosts, network congestion, or time exceeded.
- ICMP messages are encapsulated within IP datagrams, and have a specific format that consists of a type, a code, a checksum, and a data field.
- Some common ICMP message types are:
 - Echo request and echo reply: used to test the connectivity between two hosts using the ping command .
 - Destination unreachable: used to inform the sender that the destination host or network is unreachable for some reason .
 - Time exceeded: used to inform the sender that the datagram has expired in transit due to a hop limit or a timeout .
 - Parameter problem: used to inform the sender that the datagram

has an invalid or missing field in the IP header.

- Source quench: used to inform the sender that the receiver or an intermediate router is experiencing congestion and requests the sender to reduce the sending rate.
- Redirect: used to inform the sender that there is a better route to the destination and to update its routing table accordingly.
- Router advertisement and router solicitation: used to discover and advertise the presence of routers on a network.

Routing in network layer

- Routing is the process of finding the best possible path from the source host to the destination host in a network.
- Routing is performed by a special device known as a router, which works at the network layer in the OSI model and internet layer in TCP/IP model.
- A router forwards the packets based on the information available in the packet header and forwarding table.
- Forwarding is the process of moving a packet from one interface of a router to another interface based on the destination address of the packet.
- Routing and forwarding are two different functions of the network layer, but they are closely related.
- Routing algorithms are used to determine the best route for a packet, and they can be classified into two types: static and dynamic .
 - Static routing algorithms use fixed tables that are manually configured or rarely changed. They are simple, fast, and reliable, but they cannot adapt to network changes or failures.
 - Dynamic routing algorithms use protocols that automatically update the routing tables based on network conditions, such as traffic, congestion, or link failures. They are more flexible and robust, but they require more computation and communication overhead.
- Some examples of dynamic routing protocols are RIP, OSPF, EIGRP, BGP, etc.
- Routing can also be classified into two types: unicast and multicast.
 - Unicast routing is the process of sending a packet from one source to one destination. It is the most common type of routing in the internet.
 - Multicast routing is the process of sending a packet from one source to multiple destinations. It is used for applications that require efficient data distribution, such as video conferencing, online gaming, or streaming.
- Some examples of multicast routing protocols are DVMRP, PIM, MOSPF, etc.
- Routing can also be classified into two types: intra-domain and inter-domain.
 - Intra-domain routing is the process of routing packets within a single network or autonomous system (AS). It is also known as interior

gateway routing.

- Inter-domain routing is the process of routing packets between different networks or autonomous systems (ASes). It is also known as exterior gateway routing.
- Some examples of intra-domain routing protocols are RIP, OSPF, EIGRP, etc.
- Some examples of inter-domain routing protocols are BGP, EGP, etc.
- Routing is one of the main functions of the network layer, which is the third layer in the OSI model and the second layer in the TCP/IP model .
- The network layer is responsible for handling the routing and sending of data between different networks.
- The network layer also provides other services, such as packetizing, addressing, error control, congestion control, etc .
- The network layer uses protocols such as IP, ICMP, ARP, RARP, etc.
- The network layer is also known as the internet layer in the TCP/IP model.

Forwarding and Delivery in Network Layer

- The network layer supervises the handling of packets by the underlying physical networks. We define this handling as the delivery of packets to the destination.
- The delivery of a packet is called **direct** if the deliverer (host or router) and the destination are on the same network; the delivery of a packet is called **indirect** if the deliverer (host or router) and the destination are on different networks.
- The network layer also determines the route or path taken by the packets from the source to the destination. This process is called routing.
- Routing involves two basic activities: determining optimal routing paths and transporting packets through an internetwork.
- The routing process is usually performed by routers, which are network layer devices that forward packets from one network to another.
- Forwarding means placing the packet in its route to destination and it requires a routing table.
- A routing table is a data structure that stores information about the routes to various network destinations. The routing table contains a list of destination network addresses and the interface or next hop address to reach that destination.
- A router uses the destination address of a packet and the routing table to decide where to forward the packet. This is called the forwarding decision.
- Forwarding refers to the router-local action of transferring packet from an input link interface to the appropriate output link interface. Routing refers to the network-wide process that determines the end-to-end paths that packets take from source to destination.
- There are different types of routing algorithms, such as static, dynamic, distance vector, link state, hierarchical, broadcast, multicast, etc. Each al-

gorithm has its own advantages and disadvantages in terms of complexity, scalability, robustness, optimality, etc.

- Some tools or utilities that can help in packet delivery and routing are ping, traceroute, ipconfig, netstat, etc. These tools can test the connectivity, display the route, show the configuration, and monitor the traffic of a network.

Mnemonics and learning tricks

- A possible mnemonic to remember the difference between forwarding and routing is: **F**orwarding is **F**ast and **F**ocused, **R**outing is **R**easoned and **R**evised.
- A possible learning trick to understand the concept of routing is to compare it with the process of finding directions on a map. The routing algorithm is like the map, the routing table is like the list of directions, and the forwarding decision is like the choice of which direction to take at each intersection.

Static and dynamic routing in computer networks

- Static routing and dynamic routing are two methods used to determine how to send a packet toward its destination.
- Static routes are configured in advance of any network communication by the network administrator .
- Dynamic routing requires routers to exchange information with other routers to learn about paths through the network .

Key differences between static and dynamic routing

- Path selection: Static routing uses a single preconfigured route to send traffic to its destination, while dynamic routing uses complex routing algorithms to select the best route based on network conditions .
- Ability to update routes: Network administrators must manually reconfigure static routes in order to adjust routes, while dynamic routing automatically adapts to network changes .
- Routing overhead: Static routing does not generate any routing overhead, while dynamic routing consumes bandwidth and CPU resources to exchange routing information .
- Security: Static routing provides more security as it does not expose the network topology, while dynamic routing provides less security as it may be vulnerable to attacks or misconfigurations.
- Scalability: Static routing is suitable for small networks with one or two routes, while dynamic routing is suitable for large networks with many routes .
- Reliability: Static routing does not provide any backup or alternative routes in case of link failures, while dynamic routing can provide redundancy and load balancing by using multiple routes .

Examples of static and dynamic routing protocols

- Static routing does not use any specific protocol, but rather relies on manual configuration of the routing table.
- Dynamic routing uses various protocols to exchange routing information, such as:
 - Distance vector protocols: These protocols use the hop count as the metric to determine the best route. Examples are RIP and IGRP.
 - Link state protocols: These protocols use the cost of the link as the metric to determine the best route. Examples are OSPF and IS-IS.
 - Path vector protocols: These protocols use the attributes of the path as the metric to determine the best route. Examples are BGP and EIGRP.

Advantages and disadvantages of static and dynamic routing

- Static routing has the following advantages:
 - It is simple and easy to configure.
 - It does not consume any bandwidth or CPU resources.
 - It provides more security and control over the network.
- Static routing has the following disadvantages:
 - It is not scalable or adaptable to network changes.
 - It does not provide any redundancy or load balancing.
 - It requires manual intervention and maintenance.
- Dynamic routing has the following advantages:
 - It is scalable and adaptable to network changes.
 - It provides redundancy and load balancing.
 - It does not require manual intervention and maintenance.
- Dynamic routing has the following disadvantages:
 - It is complex and difficult to configure.
 - It consumes bandwidth and CPU resources.
 - It provides less security and control over the network.

Mnemonics and learning tricks for static and dynamic routing

- One possible mnemonic to remember the difference between static and dynamic routing is:
 - Static routing is **S**imple, **S**ecure, and **S**tatic.
 - Dynamic routing is **D**ifficult, **D**emanding, and **D**ynamic.
- One possible learning trick to remember the types of dynamic routing protocols is:
 - Distance vector protocols use the **D**istance to the destination as the metric.
 - Link state protocols use the **L**ink cost as the metric.
 - Path vector protocols use the **P**ath attributes as the metric.

Routing algorithms and protocols in computer networks

- Routing is the process of finding and selecting the best path for data transmission from source to destination in a network.
- Routing algorithms are the software programs that implement the logic of routing, using various metrics and criteria to determine the optimal path.
- Routing protocols are the set of rules and standards that enable routers to communicate with each other and exchange routing information.
- Routing algorithms and protocols can be classified into three major categories :
 - **Interior gateway protocols (IGPs)**: These are the routing protocols used within a single autonomous system (AS), which is a network under the same administrative control. IGPs can be further divided into two types:
 - * **Link-state routing protocols**: These protocols maintain a complete map of the network topology and calculate the shortest path to each destination using an algorithm such as Dijkstra's or Bellman-Ford. Examples of link-state protocols are Open Shortest Path First (OSPF) and Intermediate System to Intermediate System (IS-IS).
 - * **Distance-vector routing protocols**: These protocols exchange information about the distance and direction to each destination with their neighboring routers. They use an iterative process of updating their routing tables based on the information received from other routers. Examples of distance-vector protocols are Routing Information Protocol (RIP) and Interior Gateway Routing Protocol (IGRP).
 - **Exterior gateway protocols (EGPs)**: These are the routing protocols used between different autonomous systems. They are responsible for exchanging routing information and policies among different networks. The most common EGP is the Border Gateway Protocol (BGP), which is used to connect the Internet.
 - **Hybrid routing protocols**: These are the routing protocols that combine the features of both link-state and distance-vector protocols. They use a hierarchical structure to divide the network into zones or areas and use different routing algorithms within and between the zones. Examples of hybrid protocols are Enhanced Interior Gateway Routing Protocol (EIGRP) and Zone Routing Protocol (ZRP).
- Routing algorithms and protocols have various advantages and disadvantages, depending on the network size, topology, traffic, and performance requirements. Some of the factors that affect the choice of routing algorithms and protocols are :
 - **Scalability**: The ability of the routing algorithm or protocol to handle the growth of the network and the number of routers and destinations.
 - **Convergence**: The speed and efficiency of the routing algorithm or

protocol to reach a consistent and accurate state of the network after a change or failure.

- **Overhead:** The amount of bandwidth and processing power consumed by the routing algorithm or protocol to exchange and process routing information.
- **Robustness:** The ability of the routing algorithm or protocol to cope with errors, failures, and attacks in the network.
- **Flexibility:** The ability of the routing algorithm or protocol to adapt to different network conditions and requirements, such as load balancing, quality of service, and security.
- Routing algorithms and protocols are an essential component of computer networks, as they enable data to flow between different parts of the network and reach the intended destination. They are also important for the performance, reliability, and security of the network. Therefore, it is important to understand the basic concepts and principles of routing algorithms and protocols, as well as their advantages and disadvantages, for designing and managing computer networks.

Congestion control algorithms in computer networks

- Congestion control algorithms are mechanisms that control the entry of data packets into the network, enabling a better use of a shared network infrastructure and avoiding congestive collapse.
- Congestive collapse is a situation where the network performance degrades drastically due to excessive traffic and congestion.
- Congestion control algorithms can be broadly classified into two categories: open loop and closed loop.
- Open loop congestion control policies are applied to prevent congestion before it happens. They involve designing the network and choosing the appropriate protocols and parameters to avoid congestion.
- Closed loop congestion control policies are applied to detect and mitigate congestion after it happens. They involve monitoring the network state and adjusting the transmission rate or window size of the senders based on feedback signals such as packet loss, delay, or explicit notifications.
- Some examples of open loop congestion control techniques are admission control, traffic shaping, and resource reservation.
- Some examples of closed loop congestion control techniques are congestion avoidance, congestion recovery, and congestion notification.
- Congestion avoidance algorithms are implemented at the TCP layer as the mechanism to avoid congestive collapse in a network. They use packet losses and delays as signals to infer congestion and decide how fast to send data.
- Some examples of congestion avoidance algorithms are slow start, additive increase/multiplicative decrease (AIMD), congestion window (CWND), and TCP variants such as Tahoe, Reno, New Reno, Vegas, and BBR.
- Congestion recovery algorithms are implemented at the TCP layer as the

mechanism to recover from packet losses due to congestion. They use retransmission timers, acknowledgments, and fast retransmit and fast recovery techniques to resend the lost packets and resume the normal transmission.

- Congestion notification algorithms are implemented at the network layer as the mechanism to notify the senders about the congestion state of the network. They use explicit congestion notification (ECN) bits in the IP header or router-generated packets such as source quench or choke packets to inform the senders to reduce their transmission rate or window size.
- A mnemonic to remember the congestion control algorithms is: **Open Loop, Closed Loop, Avoidance, Recovery, Notification, or OLCARN**.
- A table to compare the congestion control algorithms is given below:

Algorithm	Category	Layer	Signal	Action
Admission control	Open loop	Application	N/A	Reject or accept new flows based on available resources
Traffic shaping	Open loop	Network	N/A	Control the rate and burstiness of data entering the network
Resource reservation	Open loop	Network	N/A	Reserve bandwidth and buffer space for specific flows
Slow start	Congestion avoidance	Transport	Packet loss	Increase the window size exponentially until a threshold is reached
AIMD	Congestion avoidance	Transport	Packet loss	Increase the window size linearly until a packet loss occurs, then decrease it multiplicatively

Algorithm	Category	Layer	Signal	Action
CWND	Congestion avoidance	Transport	Packet loss	Maintain a congestion window that limits the number of packets in the network
TCP variants	Congestion avoidance	Transport	Packet loss	Use different algorithms to adjust the window size based on packet loss
Retransmission timer	Congestion recovery	Transport	Timeout	Resend the lost packet after a certain time interval
Acknowledgment	Congestion recovery	Transport	Duplicate ACK	Resend the lost packet after receiving a certain number of duplicate ACKs
Fast retransmit and fast recovery	Congestion recovery	Transport	Duplicate ACK	Resend the lost packet and reduce the window size by half after receiving a certain number of duplicate ACKs
ECN	Congestion notification	Network	ECN bit	Set the ECN bit in the IP header to indicate congestion

Algorithm	Category	Layer	Signal	Action
Source quench	Congestion notification	Network	Source quench packet	Send a source quench packet to the sender to request a reduction in transmission rate
Choke packet	Congestion notification	Network	Choke packet	Send a choke packet to the sender to request a reduction in window size

IPv6 in cn

- IPv6 is the latest version of the Internet Protocol (IP), which is the set of rules that defines how devices communicate over the Internet.
- IPv6 was designed to overcome the limitations of IPv4, such as the exhaustion of address space, the lack of end-to-end connectivity, and the complexity of network configuration and management.
- IPv6 uses 128-bit addresses, which can support 2^{128} (approximately 3.4×10^{38}) possible addresses, compared to 2^{32} (approximately 4.3×10^9) addresses in IPv4.
- IPv6 also introduces new features, such as stateless address autoconfiguration, neighbor discovery, multicast, anycast, mobility, security, and transition mechanisms.
- China is one of the leading countries in IPv6 deployment and adoption, with a plan to run a single-stack IPv6 network by 2030 .
- According to the Notice on Accelerating the Large-scale Deployment and Application of IPv6 issued by the Chinese government in 2021, China aims to have 700 million active IPv6 users by 2023, plus 200 million Internet of Things devices using the protocol.
- China also plans to upgrade its core network infrastructure, backbone networks, metropolitan area networks, access networks, and content delivery networks to support IPv6.
- China's IPv6 deployment is driven by several factors, such as the demand for more IP addresses, the development of 5G and IoT technologies, the promotion of digital economy and innovation, and the alignment with global standards and trends.
- Some of the challenges and opportunities for IPv6 in China include the

coordination and cooperation among different stakeholders, the availability and affordability of IPv6 devices and applications, the security and privacy issues, and the education and awareness of users and operators.

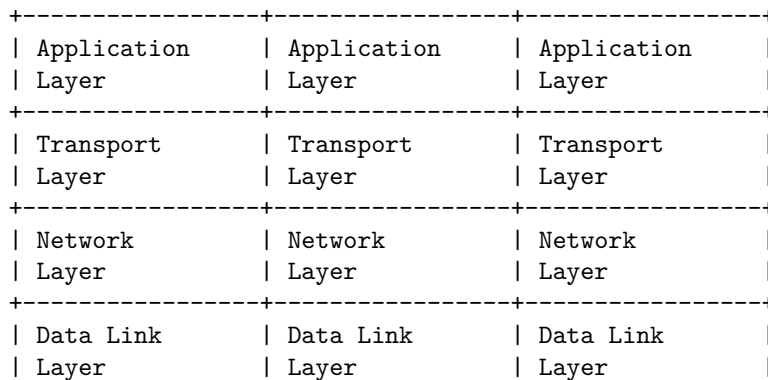
Some possible mnemonics and learning tricks for IPv6 in cn are:

- To remember the length of IPv6 addresses, think of 128 as $2^7 \times 2^7$, which is two sevens (two lucky numbers) multiplied by each other.
- To remember the format of IPv6 addresses, think of eight groups of four hexadecimal digits separated by colons, which can be abbreviated by omitting leading zeros and replacing consecutive zeros with a double colon. For example, 2001:0db8:0000:0000:ff00:0042:8329 can be written as 2001:db8::ff00:42:8329.
- To remember the types of IPv6 addresses, think of the first three bits of the address as a prefix that indicates the type. For example, 001 means global unicast, 1111 1110 10 means link-local unicast, and 1111 1111 means multicast.

Unit 4 - Transport Layer in Computer Networks

- The transport layer is a conceptual division of methods in the layered architecture of protocols in the network stack in the Internet protocol suite and the OSI model.
- The protocols of this layer provide end-to-end communication services for applications.
- The transport layer takes data from the application layer and then breaks it into smaller size segments, numbers each byte, and hands over to the network layer for delivery.
- The transport layer also reassembles the segments at the destination and delivers them to the application layer.
- The transport layer is responsible for the following functions :
 - **Addressing:** The transport layer provides the user address which is specified as a station or port. The port variable represents a specific process running on a host. The transport layer protocols need to know the port number of the application layer protocol to deliver the data to the correct process.
 - **Multiplexing and Demultiplexing:** The transport layer can multiplex multiple application layer processes on a single host by using different port numbers for each process. Similarly, the transport layer can demultiplex the incoming segments from the network layer by using the port numbers in the header and deliver them to the appropriate application layer process.
 - **Reliable Delivery:** The transport layer ensures the reliable arrival of messages across a network and provides error-checking mechanisms and data flow controls. The transport layer can detect and correct errors, retransmit lost or corrupted segments, and acknowledge the received segments. The transport layer can also use sequence numbers

- and timers to ensure the correct order and timing of the segments.
- **Flow Control:** The transport layer can regulate the amount of data that a sender can transmit to a receiver, to avoid overwhelming the receiver or the network. The transport layer can use techniques such as sliding window, stop-and-wait, or backpressure to control the flow of data.
 - **Congestion Control:** The transport layer can monitor the network conditions and adjust the rate of data transmission to avoid congestion and packet loss. The transport layer can use techniques such as additive increase multiplicative decrease (AIMD), slow start, congestion avoidance, or congestion recovery to control the congestion.
 - **Quality of Service:** The transport layer can provide different levels of service to different applications, depending on their requirements. The transport layer can use parameters such as bandwidth, delay, jitter, or reliability to specify the quality of service. The transport layer can also use techniques such as reservation, prioritization, or differentiation to allocate the network resources accordingly.
 - The transport layer can be classified into two types of protocols: connection-oriented and connectionless.
 - **Connection-oriented protocols:** These protocols establish a logical connection between the sender and the receiver before transmitting the data. The connection-oriented protocols provide reliable, ordered, and error-free delivery of data. The connection-oriented protocols use a three-way handshake to establish and terminate the connection. An example of a connection-oriented protocol is the Transmission Control Protocol (TCP).
 - **Connectionless protocols:** These protocols do not establish a logical connection between the sender and the receiver before transmitting the data. The connectionless protocols provide fast, unordered, and unreliable delivery of data. The connectionless protocols do not use any handshake or acknowledgment mechanism. An example of a connectionless protocol is the User Datagram Protocol (UDP).
 - The transport layer can be represented by the following diagram:



Physical	Physical	Physical
Layer	Layer	Layer

- The transport layer can be studied by using the following mnemonics and learning tricks:
 - To remember the functions of the transport layer, use the acronym **ARM F

Process-to-process delivery in transport layer

- The transport layer is responsible for delivering data from one process to another process on different hosts across a network .
- A process is an entity of the application layer that uses the services of the transport layer.
- The transport layer uses port numbers to identify and communicate with different processes on the same host .
- Port numbers are 16-bit integers that range from 0 to 65535.
- The transport layer can use two protocols to perform process-to-process delivery: TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) .
- TCP is a connection-oriented, reliable, and full-duplex protocol that provides flow control, congestion control, and error recovery .
- UDP is a connectionless, unreliable, and simple protocol that does not provide any of these features .
- TCP and UDP use different header formats to encapsulate the data from the application layer .
- TCP header contains fields such as source port, destination port, sequence number, acknowledgment number, flags, window size, checksum, and urgent pointer .
- UDP header contains only four fields: source port, destination port, length, and checksum .
- The transport layer segments the data from the application layer and adds a header to each segment before passing it to the network layer .
- The network layer encapsulates the segments into packets (or datagrams) and adds a header to each packet before passing it to the data link layer .
- The data link layer encapsulates the packets into frames and adds a header and a trailer to each frame before passing it to the physical layer .
- The physical layer converts the frames into bits and transmits them over the physical medium .
- The reverse process happens at the receiver side, where the transport layer receives the segments from the network layer, checks the port numbers, and delivers the data to the corresponding process .

Figure 1 shows an example of process-to-process delivery using TCP and UDP.

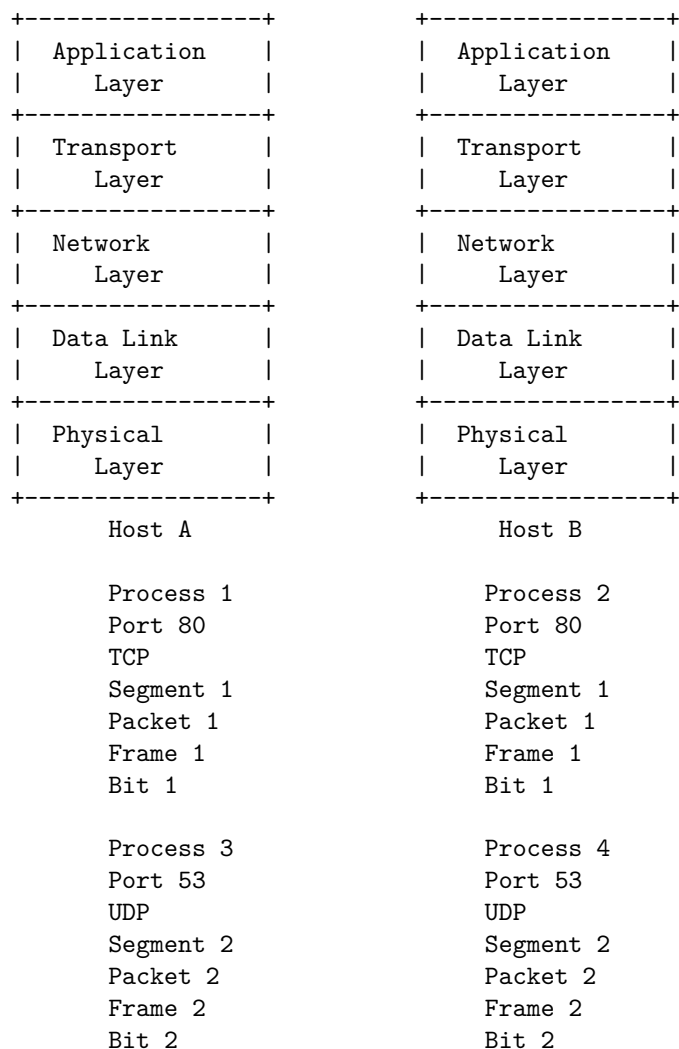


Figure 1: Process-to-process delivery using TCP and UDP

Some mnemonics and learning tricks for process-to-process delivery in transport layer are:

- Remember that port numbers are 16-bit integers by thinking of PORT as an acronym for Power Of 2 Times 16.
- Remember that TCP is connection-oriented, reliable, and full-duplex by thinking of TCP as an acronym for Trust

Transport layer protocols

- Transport layer protocols are responsible for providing end-to-end communication services for applications over a network.
- Transport layer protocols lie between the user applications and the network layer, and use the network layer protocols (such as IP) to send and receive packets.
- Transport layer protocols can offer different types of services, such as reliable or unreliable, connection-oriented or connectionless, stream-oriented or message-oriented, etc.
- The most common transport layer protocols in the Internet are Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) .
- TCP is a reliable, connection-oriented, stream-oriented protocol that ensures the correct delivery of packets in the same order as they were sent . TCP uses mechanisms such as sequence numbers, acknowledgments, timers, and retransmissions to deal with packet loss, duplication, corruption, and reordering. TCP also provides flow control and congestion control to regulate the rate of data transmission and avoid network congestion.
- UDP is an unreliable, connectionless, message-oriented protocol that does not guarantee the delivery or ordering of packets . UDP does not use any mechanisms to recover from packet loss, duplication, corruption, or reordering. UDP also does not provide any flow control or congestion control, and relies on the application layer to handle these issues. UDP is suitable for applications that require low latency, high throughput, or real-time communication, such as voice over IP, video streaming, online gaming, etc.
- Other transport layer protocols that have been defined and implemented include the Datagram Congestion Control Protocol (DCCP) and the Stream Control Transmission Protocol (SCTP).
- DCCP is a connection-oriented, message-oriented protocol that provides unreliable delivery with congestion control. DCCP is designed for applications that can tolerate packet loss, but need to avoid network congestion, such as multimedia streaming, telephony, etc.
- SCTP is a connection-oriented, message-oriented protocol that provides reliable delivery with multiple streams and multihoming. SCTP allows multiple messages to be sent and received concurrently over different streams within a single connection, and also supports multiple network addresses for each endpoint. SCTP is designed for applications that need high availability, fault tolerance, and load balancing, such as web servers, telecommunication systems, etc.

Some possible mnemonics and learning tricks for the transport layer protocols are:

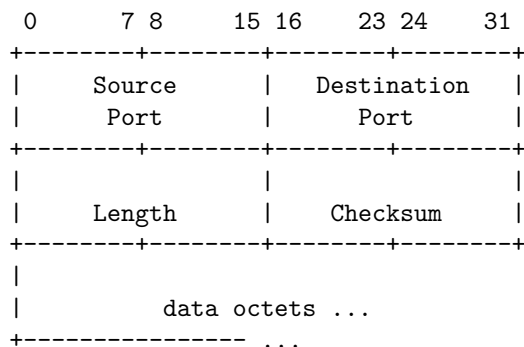
- TCP: **T**ransport **C**ontrol **P**rotocol -> **T**rustworthy, **C**onected, **P**ersistent
- UDP: **U**ser **D**atagram **P**rotocol -> **U**nreliable, **D**isconnected,

Performance

- DCCP: **D**atagram **C**ongestion **C**ontrol **P**rotocol -> **D**rop **C**apable, **C**ongestion **C**autious, **P**acketized
- SCTP: **S**tream **C**ontrol **T**ransmission **P**rotocol -> **S**imultaneous, **C**onected, **T**ransportable, **P**lural

UDP Transport layer protocol

- UDP stands for User Datagram Protocol. It is a transport layer protocol that is part of the Internet Protocol suite, also known as UDP/IP suite .
- UDP is a simple, unreliable and connectionless protocol. It does not establish a connection before sending data, nor does it guarantee that the data will be delivered or received in order or without errors .
- UDP adds a 16-byte header to the data from the upper layer. The header contains four fields: source port, destination port, length and checksum. The source and destination ports identify the endpoints of the communication. The length field specifies the total length of the UDP datagram, including the header and the data. The checksum field provides a basic error detection mechanism by verifying the integrity of the header and the data .



- UDP is suitable for applications that require speed, simplicity or low overhead, such as real-time audio or video streaming, online gaming, DNS queries, etc. UDP is also used for multicasting and broadcasting, where a single sender can transmit data to multiple receivers .
- UDP has some disadvantages, such as lack of reliability, congestion control, flow control, error recovery, etc. UDP applications have to implement these features themselves if needed. UDP also has a limited payload size of 65,507 bytes, which is the maximum length of the UDP datagram minus the header size .
- A mnemonic to remember the four fields of the UDP header is: **S**ource **P**ort, **D**estination **P**ort, **L**ength, **C**hecksum, or **SPDLC**. Another mnemonic is: **S**end **P**ackets **D**irectly **L**ike **C**razy, or **SPDLC**.

TCP Transport layer protocol

- TCP stands for Transmission Control Protocol. It is a transport layer protocol that facilitates the transmission of packets from source to destination.
- TCP is a connection-oriented protocol that means it establishes the connection prior to the communication that occurs between the computing devices in a network.
- TCP is a reliable protocol as it follows the flow and error control mechanism. It also supports the acknowledgment mechanism, which checks the state and sound arrival of the data.
- TCP uses a three-way handshake to establish a connection between the sender and the receiver. The three steps are: SYN, SYN-ACK, and ACK .
- TCP uses a sliding window protocol to control the flow of data. The sender maintains a window of packets that it can send without waiting for an acknowledgment. The receiver maintains a window of packets that it can receive and acknowledge. The size of the window can vary depending on the network conditions .
- TCP uses a checksum to detect errors in the packets. The checksum is a value calculated from the data in the packet. The sender adds the checksum to the packet header and the receiver verifies it. If the checksum does not match, the receiver discards the packet and requests a retransmission .
- TCP uses a sequence number and an acknowledgment number to identify and acknowledge the packets. The sequence number is the number of the first byte in the packet. The acknowledgment number is the number of the next expected byte. The sender and the receiver use these numbers to keep track of the data transfer .
- TCP uses a timer to handle timeouts and retransmissions. The timer is set when a packet is sent and reset when an acknowledgment is received. If the timer expires before an acknowledgment is received, the sender assumes that the packet is lost and retransmits it .
- TCP uses a congestion control mechanism to avoid overloading the network. The congestion control mechanism adjusts the window size and the transmission rate based on the feedback from the network. The feedback can be implicit, such as packet loss or delay, or explicit, such as congestion notification messages .
- TCP supports various options and features, such as urgent data, selective acknowledgments, window scaling, timestamps, and multipath TCP .
- TCP is widely used for applications that require reliable and ordered delivery of data, such as web browsing, email, file transfer, and remote login .

Multiplexing in transport layer

Multiplexing in transport layer is the process of collecting the data from multiple application processes of the sender, enveloping that data with headers and sending them as a whole to the intended receiver. Multiplexing in transport layer extends the host-to-host delivery service provided by the network layer to a process-to-process delivery service for applications running on the hosts.

Some points to remember about multiplexing in transport layer are:

- Multiplexing in transport layer requires that sockets have unique identifiers, and each segment have special fields that indicate the sockets to which the segment is to be delivered.
- Multiplexing in transport layer can be connectionless or connection-oriented, depending on the transport protocol used.
- Connectionless multiplexing uses port numbers to identify the sockets, and includes the source and destination port numbers in the segment header.
- Connection-oriented multiplexing uses a combination of port numbers and socket numbers to identify the sockets, and includes the source and destination socket numbers in the segment header.
- Multiplexing in transport layer allows multiple applications to share the same network resources and communicate with different destinations simultaneously.
- Multiplexing in transport layer is performed by the transport layer protocols, such as UDP and TCP.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have asked me to write on the topic of connection management in transport layer. Here is my response:

Connection management in transport layer

- The transport layer is responsible for providing end-to-end communication between applications on different hosts in a network.
- The transport layer can use either connection-oriented or connectionless protocols to deliver data between applications.
- Connection-oriented protocols establish a logical connection between the sender and the receiver before exchanging data. They also maintain the connection state and ensure reliable and ordered delivery of data. An example of a connection-oriented protocol is TCP (Transmission Control Protocol).
- Connectionless protocols do not establish or maintain any connection state between the sender and the receiver. They simply send data as independent packets, without guaranteeing reliability or order. An example of a connectionless protocol is UDP (User Datagram Protocol).
- Connection management in transport layer involves the following steps for connection-oriented protocols:

- Connection establishment: The sender and the receiver agree on the parameters and rules for the data transfer, such as the initial sequence numbers, window sizes, and timeout values. This is done by exchanging control packets, such as SYN, SYN-ACK, and ACK, in a three-way handshake.
- Data transfer: The sender and the receiver exchange data packets, each with a sequence number and an acknowledgment number. The sender uses a sliding window to control the amount of data sent without acknowledgment. The receiver uses a sliding window to control the amount of data received out of order. The sender and the receiver also use timers and retransmission mechanisms to handle packet loss and errors.
- Connection termination: The sender and the receiver close the connection by exchanging control packets, such as FIN, FIN-ACK, and ACK, in a four-way handshake. The connection state is released and the resources are freed.
- Connection management in transport layer involves the following steps for connectionless protocols:
 - There is no connection establishment or termination for connectionless protocols. The sender and the receiver do not need to agree on any parameters or rules for the data transfer.
 - Data transfer: The sender and the receiver exchange data packets, each with a source and destination port number. The sender does not keep track of the packets sent or received. The receiver does not keep track of the packets received or expected. There is no guarantee of reliability or order for the data packets.
- Some mnemonics and learning tricks for connection management in transport layer are:
 - To remember the steps of connection establishment and termination for connection-oriented protocols, use the acronym SSTF: SYN, SYN-ACK, Transfer, FIN.
 - To remember the difference between connection-oriented and connectionless protocols, use the analogy of a phone call and a postcard. A phone call is like a connection-oriented protocol, where you need to dial a number, wait for an answer, talk, and hang up. A postcard is like a connectionless protocol, where you just write a message, put a stamp, and send it.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have asked me to write on the topic of flow control in transport layer. Here is my response:

Flow control in transport layer

- Flow control is a mechanism that regulates the amount of data that a sender can transmit to a receiver, without overwhelming the receiver's

buffer capacity.

- Flow control is necessary because the sender and the receiver may have different processing speeds, network bandwidths, or buffer sizes, which can cause congestion or data loss if the sender sends too much data too fast.
- Flow control can be implemented at different layers of the network stack, such as the data link layer, the transport layer, or the application layer. In this response, we will focus on the transport layer, which is responsible for end-to-end communication between two processes on different hosts.
- The transport layer can use two main methods for flow control: stop-and-wait and sliding window.

Stop-and-wait

- Stop-and-wait is a simple method of flow control, where the sender sends one data unit (such as a packet or a segment) at a time, and waits for an acknowledgment (ACK) from the receiver before sending the next one.
- The sender maintains a timer for each data unit, and if the timer expires before receiving an ACK, the sender assumes that the data unit was lost or corrupted, and retransmits it.
- The receiver sends an ACK for each data unit it receives correctly, and discards any duplicate or out-of-order data units.
- The advantage of stop-and-wait is that it is easy to implement and avoids congestion, since the sender can only have one outstanding data unit at a time.
- The disadvantage of stop-and-wait is that it is inefficient, since the sender has to wait for an ACK after each data unit, which introduces a lot of idle time and reduces the throughput. The throughput is limited by the round-trip time (RTT), which is the time it takes for a data unit to travel from the sender to the receiver and back.

Sliding window

- Sliding window is a more advanced method of flow control, where the sender can send multiple data units without waiting for an ACK, as long as the number of outstanding data units does not exceed a certain limit, called the window size.
- The sender maintains a window, which is a range of sequence numbers that indicate the data units that have been sent but not yet acknowledged. The sender can send any data unit whose sequence number falls within the window, and updates the window as it receives ACKs from the receiver.
- The receiver also maintains a window, which is a range of sequence numbers that indicate the data units that it expects to receive next. The receiver can accept any data unit whose sequence number falls within the window, and sends an ACK for the highest sequence number it has received correctly. The receiver updates the window as it receives new data units, and may use a buffer to store out-of-order data units until they can

be delivered to the application.

- The advantage of sliding window is that it improves the efficiency and throughput of the communication, since the sender can utilize the available bandwidth more effectively, and the receiver can handle out-of-order data units more gracefully.
- The disadvantage of sliding window is that it is more complex to implement and requires more coordination between the sender and the receiver, such as agreeing on the window size and handling errors and retransmissions.

Mnemonics and learning tricks

- To remember the difference between stop-and-wait and sliding window, you can use the following mnemonics:
 - Stop-and-wait: One by one, slow and done.
 - Sliding window: Many by many, fast and canny.
- To remember the formula for the maximum throughput of stop-and-wait, you can use the following trick:
 - The maximum throughput is equal to the data unit size divided by the RTT, or $T = S / \text{RTT}$.
 - You can think of this as: Time = Size / Round-Trip-Time, or $T = S / \text{RTT}$.

Transmission in Transport Layer

The transport layer is the fourth layer of the OSI model, which provides end-to-end communication between application processes running on different hosts within a network. The transport layer performs the following functions:

- It takes the data from the upper layer (application layer), and it divides the data into smaller packets, called segments, and then transmits them to the network layer (layer 3).
- It assigns a port number to each segment, which identifies the source and destination application processes. The port number is a 16-bit number that ranges from 0 to 65535. The port number is added to the segment header, along with other information such as sequence number, acknowledgment number, checksum, etc.
- It ensures reliable transmission of segments by using mechanisms such as flow control, error control, and congestion control. Flow control regulates the rate of data transmission between the sender and the receiver, to avoid buffer overflow. Error control detects and corrects errors in the segments, by using techniques such as checksum, acknowledgment, and retransmission. Congestion control prevents network congestion by adjusting the window size or the sending rate of the sender, based on the feedback from the network or the receiver.
- It provides two modes of transmission: connection-oriented and connectionless. Connection-oriented transmission establishes a logical connection

between the sender and the receiver before sending the data, and maintains the connection until the data transfer is complete. Connectionless transmission does not require a connection establishment or termination, and sends the data as independent datagrams. Connection-oriented transmission is more reliable and ordered, but also more complex and overhead. Connectionless transmission is simpler and faster, but also less reliable and unordered.

The most common transport protocols of TCP/IP are the Transmission Control Protocol (TCP) and the User Datagram Protocol (UDP). TCP is a connection-oriented protocol that provides reliable, ordered, and error-free transmission of segments. UDP is a connectionless protocol that provides fast, simple, and unreliable transmission of datagrams. TCP is used for applications that require high reliability and accuracy, such as web browsing, email, file transfer, etc. UDP is used for applications that can tolerate some loss and delay, such as video streaming, voice over IP, online gaming, etc.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have asked me to write on the topic of window management in transport layer. Here is my response:

Window management in transport layer

- Window management is a technique used by the transport layer protocols, such as TCP and UDP, to control the flow of data between the sender and the receiver.
- Window management ensures that the sender does not overwhelm the receiver or the network with too many packets at once, and that the receiver can acknowledge the received packets and request retransmission of lost or corrupted packets.
- Window management also helps to improve the efficiency and reliability of data transmission by adapting to the changing network conditions, such as congestion, delay, and errors.
- Window management involves two main concepts: window size and sliding window.
- Window size is the number of packets or bytes that the sender can send or the receiver can accept at a given time. Window size can be fixed or variable, depending on the protocol and the network situation.
- Sliding window is a technique that allows the sender and the receiver to keep track of the sequence of packets that are sent and received, and to update the window size accordingly. Sliding window can be implemented in different ways, such as stop-and-wait, go-back-N, and selective repeat.
- Stop-and-wait is the simplest sliding window technique, where the sender sends one packet at a time and waits for an acknowledgment from the

receiver before sending the next packet. The window size is one packet in this case. Stop-and-wait is easy to implement, but it is inefficient and slow, especially for long-distance or noisy networks.

- Go-back-N is a sliding window technique where the sender can send multiple packets without waiting for acknowledgments, but the receiver can only send cumulative acknowledgments for the last packet received in order. The window size is N packets in this case. Go-back-N is faster and more efficient than stop-and-wait, but it has a drawback of wasting bandwidth and resources by retransmitting all the packets after a lost or corrupted packet, even if some of them were received correctly by the receiver.
- Selective repeat is a sliding window technique where the sender can send multiple packets without waiting for acknowledgments, and the receiver can send individual acknowledgments for each packet received, regardless of the order. The window size is N packets in this case. Selective repeat is the most efficient and reliable sliding window technique, but it is also the most complex and requires more memory and processing power at both ends.
- A mnemonic to remember the three sliding window techniques is: Stop, Go, Select (S-G-S).
- A learning trick to understand the sliding window technique is to imagine a conveyor belt with N slots, where each slot can hold one packet. The sender puts packets on the conveyor belt and moves it forward, while the receiver takes packets off the conveyor belt and moves it backward. The window size is the number of slots on the conveyor belt, and the sliding window is the movement of the conveyor belt. The different sliding window techniques can be visualized as follows:
- Stop-and-wait: The sender puts one packet on the first slot and waits for the receiver to take it off and send an acknowledgment. Then the sender puts another packet on the first slot and repeats the process. The conveyor belt does not move in this case.
- Go-back-N: The sender puts packets on the slots from the first to the Nth slot and sends them all at once. The receiver takes packets off the slots in order and sends a cumulative acknowledgment for the last packet received. If the receiver detects a lost or corrupted packet, it discards all the subsequent packets and sends a negative acknowledgment. The sender then retransmits all the packets from the lost or corrupted packet to the Nth packet. The conveyor belt moves forward and backward in this case.
- Selective repeat: The sender puts packets on the slots from the first to the Nth slot and sends them all at once. The receiver takes packets off the slots in any order and sends an individual acknowledgment for each packet received. If the receiver detects a lost or corrupted packet, it requests

a retransmission of that packet only. The sender then retransmits the requested packet only. The conveyor belt moves forward and backward in this case, but with more flexibility and efficiency.

TCP Congestion Control in Transport Layer

- TCP stands for Transmission Control Protocol. It is a connection-oriented and reliable transport layer protocol that facilitates the transmission of packets from source to destination.
- TCP congestion control is the main technique that prevents the Internet from slowing down due to over-utilization of network resources. It is the process of adjusting the sending rate of TCP packets based on the network conditions and feedback signals.
- TCP congestion control has three main objectives:
 - Efficiency: To utilize the network capacity as much as possible without causing congestion.
 - Fairness: To allocate the network resources fairly among competing TCP flows.
 - Stability: To avoid oscillations in the sending rate and maintain a smooth traffic pattern.
- TCP congestion control consists of four main components:
 - Congestion window (cwnd): The amount of data that the sender can transmit without receiving an acknowledgment from the receiver. It is dynamically adjusted by the sender based on the network conditions and feedback signals.
 - Slow start threshold (ssthresh): The threshold that determines the mode of congestion control. If cwnd is below ssthresh, the sender is in slow start mode and increases cwnd exponentially. If cwnd is above ssthresh, the sender is in congestion avoidance mode and increases cwnd linearly.
 - Acknowledgment (ACK): The message that the receiver sends to the sender to confirm the receipt of a packet or a group of packets. It also indicates the next expected sequence number from the sender.
 - Congestion signal: The signal that indicates the presence of congestion in the network. It can be either an explicit signal, such as an Explicit Congestion Notification (ECN) bit in the packet header, or an implicit signal, such as a packet loss or a timeout event.
- TCP congestion control has three main phases:
 - Slow start: The initial phase of congestion control, where the sender starts with a small cwnd and increases it exponentially for every ACK received, until it reaches ssthresh or a congestion signal occurs.
 - Congestion avoidance: The normal phase of congestion control, where the sender increases cwnd linearly for every ACK received, until a congestion signal occurs.
 - Congestion detection: The recovery phase of congestion control, where the sender detects a congestion signal and reduces cwnd and

ssthresh accordingly, and then resumes slow start or congestion avoidance mode.

- TCP congestion control uses different algorithms to implement the above components and phases, such as Reno, NewReno, Tahoe, Vegas, Cubic, BBR, etc. Each algorithm has its own advantages and disadvantages in terms of performance, fairness, and stability .
- TCP congestion control is a complex and evolving topic that requires a deep understanding of the network dynamics and the trade-offs involved. It is also influenced by various factors, such as the operating system, the application, the router, the link, etc.

: <http://witestlab.poly.edu/blog/tcp-congestion-control-basics/>
<https://www.javatpoint.com/tcp>
<https://www.geeksforgeeks.org/tcp-congestion-control/>
<https://www.baeldung.com/cs/tcp-flow-control-vs-congestion-control>
<https://networkengineering.stackexchange.com/questions/69932/congestion-control-in-transport-and-network-layers-in-tcp-ip>

Quality of service in transport layer

- Quality of service (QoS) is the use of mechanisms or technologies that work on a network to control traffic and ensure the performance of critical applications with limited network capacity.
- The transport layer is responsible for enhancing the QoS provided by the network layer by offering reliable and efficient end-to-end data transmission between application processes.
- The transport layer can provide QoS in terms of:
 - Throughput: the amount of data transferred per unit time.
 - Delay: the time taken for a packet to travel from the source to the destination.
 - Jitter: the variation in delay of packets.
 - Reliability: the probability of successful delivery of packets.
 - Security: the protection of data from unauthorized access or modification.
- The transport layer can use different protocols and techniques to achieve QoS, such as:
 - TCP: a connection-oriented, reliable, and congestion-controlled protocol that uses acknowledgments, retransmissions, and window-based flow control to ensure data integrity and efficiency.
 - UDP: a connectionless, unreliable, and best-effort protocol that does not use any error or flow control mechanisms and is suitable for real-time applications that can tolerate some packet loss or delay.
 - SCTP: a connection-oriented, reliable, and message-oriented protocol that supports multiple streams, multihoming, and partial reliability to provide flexibility and resilience for applications that require both reliability and timeliness.

- RTP: a protocol that works on top of UDP and provides end-to-end delivery services for real-time applications that involve audio and video data, such as VoIP and video conferencing. RTP provides features such as sequence numbers, timestamps, payload type identification, and source identification to enable synchronization, jitter compensation, and quality monitoring.
- RSVP: a protocol that works on both the network and the transport layer and enables applications to reserve network resources along a path from the source to the destination to guarantee QoS parameters such as bandwidth, delay, and reliability. RSVP uses signaling messages to establish, maintain, and tear down reservations and to communicate QoS requirements and availability.
- The transport layer can also cooperate with other layers to achieve QoS, such as:
 - The application layer: the transport layer can use the information provided by the application layer about the QoS requirements and preferences of the application to select the appropriate protocol and parameters for data transmission.
 - The network layer: the transport layer can use the feedback provided by the network layer about the network conditions and congestion to adjust the transmission rate and window size of the transport protocol to avoid congestion and packet loss.
 - The link layer: the transport layer can use the services provided by the link layer to ensure reliable and efficient data transmission over the physical medium, such as error detection and correction, framing, and medium access control.

Unit 5 - Application Layer in Computer Networks

- The application layer is the highest layer in the protocol suite. It provides services to the user and interacts with network applications .
- Examples of network applications are web browsers, chat clients, email clients, file transfer, remote login, etc .
- The application layer uses a logical connection, which means that the two application layers assume that there is an imaginary direct connection between them.
- The application layer can use different paradigms for communication, such as client-server, peer-to-peer, or hybrid.
- Client-server paradigm: The service provider is an application program called the server process, and the service requester is another application program called the client process. The server is usually passive and waits for requests from clients, while the client is usually active and initiates the communication. Examples of client-server applications are HTTP, FTP, SMTP, etc .

- Peer-to-peer paradigm: The service provider and the service requester are both application programs called peers. There is no distinction between clients and servers, and each peer can act as both. The peers are usually transient and dynamic, and they communicate directly with each other without a central server. Examples of peer-to-peer applications are BitTorrent, Skype, etc .
- Hybrid paradigm: The service provider and the service requester are both application programs that can act as both clients and servers, but they also use a central server for some functions, such as coordination, authentication, or indexing. Examples of hybrid applications are Napster, Spotify, etc.
- Some of the common protocols and services in the application layer are:
 - Domain Name System (DNS): A distributed database system that maps domain names to IP addresses and provides other information about hosts on the Internet .
 - Simple Network Management Protocol (SNMP): A protocol that allows network administrators to monitor and control network devices and resources .
 - Electronic Mail (Email): A service that allows users to send and receive messages over the Internet . The main protocols involved in email are Simple Mail Transfer Protocol (SMTP), Post Office Protocol (POP), and Internet Message Access Protocol (IMAP).
 - World Wide Web (WWW): A service that allows users to access and share information over the Internet using hypertext documents and links . The main protocol involved in WWW is Hypertext Transfer Protocol (HTTP).
 - Streaming Audio and Video: A service that allows users to play multimedia content over the Internet without downloading the entire file . The main protocols involved in streaming are Real-Time Streaming Protocol (RTSP), Real-Time Transport Protocol (RTP), and Real-Time Control Protocol (RTCP).
- Some of the challenges and design issues in the application layer are:
 - Scalability: The application layer should be able to handle a large number of users and requests without degrading the performance or quality of service .
 - Security: The application layer should be able to protect the data and the users from unauthorized access, modification, or disclosure . Some of the security mechanisms are encryption, authentication, authorization, and digital signatures.
 - Reliability: The application layer should be able to handle errors and failures in the network and ensure that the data is delivered correctly and completely . Some of the reliability mechanisms are acknowledgments, retransmissions, timeouts, and checksums.

- Interoperability: The application layer should be able to communicate with different types of devices and platforms using common standards and formats . Some of the interoperability mechanisms are application programming interfaces (APIs), data serialization, and data compression.
- A possible mnemonic to remember the main functions of the application layer is:
 - **A**ccess and share information
 - **P**rovide services to the user
 - **P**rotocols and paradigms for communication
 - **L**ogical connection between applications
 - **I**nteract with network applications
 - **C**hallenges and design issues
 -

Domain Name System

- The Domain Name System (DNS) is a hierarchical and distributed naming system for computers, services, and other resources in the Internet or other Internet Protocol (IP) networks.
- DNS is the main index of the internet that directs traffic for queries across the web. It maps the name people use to locate a website to the IP address that a computer uses to locate that website.
- DNS is an internet service that translates the domain name into IP addresses. For example, the domain name `google.com` is translated to the IP address `142.250.64.78`.
- DNS works by using a network of servers called DNS servers. Each DNS server stores a part of the DNS database, called a zone, that contains information about a set of domain names and their corresponding IP addresses.
- When a user requests a domain name, such as `google.com`, the user's device contacts a DNS server, usually provided by the user's internet service provider (ISP). This DNS server is called a recursive resolver, because it recursively queries other DNS servers to find the answer.
- The recursive resolver first checks its own cache to see if it already knows the IP address for the domain name. If not, it sends a query to a root server, which is one of the 13 servers that store information about the top-level domains (TLDs), such as `.com`, `.org`, `.net`, etc.
- The root server responds with a referral to the authoritative server for the TLD, such as `.com`. The recursive resolver then sends a query to the authoritative server, which responds with a referral to the authoritative server for the second-level domain, such as `google.com`.

- Online games, such as Fortnite, Minecraft, Call of Duty, etc., that use DNS to connect players to game servers
- Streaming services, such as Netflix, YouTube, Spotify, etc., that use DNS to deliver content from media servers
- Cloud services, such as AWS, Azure, Google Cloud, etc., that use DNS to provide various services and resources to customers

World Wide Web

- The World Wide Web (WWW) or the Web is a part of the Internet that allows users to access and view documents and other resources using a web browser and a graphical user interface (GUI) .
- The Web was invented by Tim Berners-Lee in 1989 at CERN, the European Organization for Nuclear Research .
- The Web is based on the concept of hypertext, which is a way of linking and presenting information using text and graphics that can be clicked to access other documents or resources .
- The Web uses a set of protocols and standards to communicate and exchange information, such as:
 - HTTP (Hypertext Transfer Protocol): the protocol that defines how web browsers and web servers communicate and request and deliver web pages .
 - HTML (Hypertext Markup Language): the language that defines the structure and content of web pages using tags and attributes .
 - URL (Uniform Resource Locator): the address that identifies the location and name of a web resource, such as a web page, an image, or a video .
 - CSS (Cascading Style Sheets): the language that defines the style and appearance of web pages, such as fonts, colors, and layouts .
 - JavaScript: the scripting language that adds interactivity and functionality to web pages, such as animations, forms, and validations .
- The Web is composed of billions of web pages that are stored on web servers, which are computers that host and deliver web resources over the Internet .
- The Web is accessed by web browsers, which are software applications that allow users to view and interact with web pages and other web resources .
- The Web is organized by web sites, which are collections of related web pages that share a common domain name, such as www.bing.com .
- The Web is searched by web search engines, which are web sites that allow users to find and retrieve web pages and other web resources based on keywords or queries, such as Bing, Google, or Yahoo .

Some mnemonics and learning tricks for the World Wide Web are:

- WWW stands for World Wide Web, which can be remembered as “W3” or “3W”.

- HTTP stands for Hypertext Transfer Protocol, which can be remembered as “Hi, To The Page”.
- HTML stands for Hypertext Markup Language, which can be remembered as “How To Make Links”.
- URL stands for Uniform Resource Locator, which can be remembered as “U R Lucky”.
- CSS stands for Cascading Style Sheets, which can be remembered as “Cool Styling Stuff”.
- JavaScript can be remembered as “Java + Script”, where Java is a programming language and Script is a written text.

Hyper Text Transfer Protocol

- Hyper Text Transfer Protocol (HTTP) is an application-layer protocol for transmitting hypermedia documents, such as HTML .
- HTTP is the underlying protocol used by the World Wide Web, developed by Tim Berners-Lee .
- HTTP defines how messages are formatted and transmitted, and what actions Web servers and browsers should take in response to various commands .
- HTTP is a stateless protocol, meaning that each request is independent of the previous one and the server does not keep track of the client’s state .
- HTTP uses a client-server model, where the client initiates a request and the server responds with a response .
- HTTP requests and responses consist of a start-line, zero or more headers, an empty line, and an optional message body .
- The start-line of a request contains the method, the URI, and the HTTP version, such as `GET /index.html HTTP/1.1` .
- The start-line of a response contains the HTTP version, the status code, and the reason phrase, such as `HTTP/1.1 200 OK` .
- The headers provide additional information about the request or the response, such as the content type, the content length, the date, the server name, etc .
- The message body contains the actual data of the request or the response, such as the HTML document, the image, the JSON data, etc .
- HTTP supports different methods for different purposes, such as GET, POST, PUT, DELETE, HEAD, OPTIONS, etc .
- HTTP supports different status codes for different outcomes, such as 200 for success, 404 for not found, 500 for internal server error, etc .
- HTTP can be extended with new headers, methods, status codes, and media types, as long as they follow the syntax and semantics of the protocol .
- HTTP can be secured with HTTPS, which uses SSL/TLS to encrypt the communication between the client and the server .

Mnemonics and learning tricks

- To remember the structure of an HTTP request or response, you can use the acronym SLHM (Start-line, Headers, empty Line, Message body).
- To remember the common HTTP methods, you can use the acronym GPOD (GET, POST, PUT, DELETE).
- To remember the common HTTP status codes, you can use the following phrases:
 - 1xx: Informational (I need more information)
 - 2xx: Success (You did it!)
 - 3xx: Redirection (Go somewhere else)
 - 4xx: Client error (You messed up)
 - 5xx: Server error (I messed up)

Examples

- An example of an HTTP request:

```
GET /index.html HTTP/1.1
Host: www.example.com
User-Agent: Mozilla/5.0
Accept: text/html
```

- An example of an HTTP response:

```
HTTP/1.1 200 OK
Date: Wed, 15 Mar 2023 13:55:49 GMT
Server: Apache
Content-Type: text/html
Content-Length: 1234
```

```
<html>
<head>
<title>Example Page</title>
</head>
<body>
<h1>Hello, World!</h1>
</body>
</html>
```

Advantages

- HTTP is simple and easy to implement .
- HTTP is flexible and extensible .
- HTTP is widely supported by browsers, servers, and proxies .
- HTTP is compatible with different media types and formats .

Disadvantages

- HTTP is stateless, meaning that it cannot maintain the context or the history of the communication .
- HTTP is insecure, meaning that it can be intercepted, modified, or spoofed by attackers .
- HTTP is inefficient, meaning that

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have asked me to write on the topic of electronic mail in application layer. Here is my response:

Electronic mail in application layer

- Electronic mail (email) is a widely used application that allows users to exchange messages across the Internet or other networks.
- Email has three major components: user agents, mail servers, and simple mail transfer protocol (SMTP).
- User agents are programs that allow users to read, compose, send, and receive email messages. Examples of user agents are Outlook, Gmail, Thunderbird, etc.
- Mail servers are computers that store and forward email messages. Each mail server has a mailbox for each user, which is a collection of messages that have been sent to that user. Mail servers also run a mail transfer agent (MTA), which is a program that sends and receives messages using SMTP.
- SMTP is an application layer protocol that defines how email messages are formatted, addressed, and transferred between mail servers. SMTP uses TCP as the transport layer protocol and port 25 as the well-known port number. SMTP has two phases: a handshaking phase and a message transfer phase. In the handshaking phase, the sender and the receiver mail servers exchange greetings and identify themselves. In the message transfer phase, the sender mail server sends one or more email messages to the receiver mail server, followed by a quit command to end the connection.
- An email message consists of two parts: a header and a body. The header contains information such as the sender, the receiver, the subject, the date, etc. The body contains the actual content of the message, which can be text, images, attachments, etc.
- An email address has the form of user@domain, where user is the name of the user and domain is the name of the mail server. For example, alice@example.com is an email address for a user named Alice at a mail server named example.com.
- To send an email message from one user to another, the following steps are involved:

1. The sender user agent creates the message and specifies the sender and the receiver email addresses.
 2. The sender user agent contacts the sender mail server and sends the message using SMTP.
 3. The sender mail server looks up the domain name of the receiver email address in the DNS and obtains the IP address of the receiver mail server.
 4. The sender mail server contacts the receiver mail server and sends the message using SMTP.
 5. The receiver mail server stores the message in the mailbox of the receiver user.
 6. The receiver user agent contacts the receiver mail server and retrieves the message using a mail access protocol, such as POP3, IMAP, or HTTP.
- A mail access protocol is an application layer protocol that allows a user agent to access and manipulate the messages stored in a mail server. There are three common mail access protocols: post office protocol version 3 (POP3), internet message access protocol (IMAP), and hypertext transfer protocol (HTTP).
 - POP3 is a simple protocol that allows a user agent to download all the messages from a mail server and delete them from the server. POP3 uses TCP as the transport layer protocol and port 110 as the well-known port number. POP3 has three phases: authorization, transaction, and update. In the authorization phase, the user agent sends the username and password to the mail server and receives a positive or negative response. In the transaction phase, the user agent issues commands to list, retrieve, and delete the messages from the mail server. In the update phase, the mail server deletes the messages that have been marked for deletion and closes the connection.
 - IMAP is a more complex protocol that allows a user agent to access and manipulate the messages from a mail server without downloading them. IMAP uses TCP as the transport layer protocol and port 143 as the well-known port number. IMAP has two modes: connected and disconnected. In the connected mode, the user agent maintains a connection with the mail server and can issue commands to list, select, fetch, store, copy, and delete the messages from the mail server. In the disconnected mode, the user agent can work offline with the messages that have been downloaded and synchronize them with the mail server later. IMAP also supports folders, flags, and searches for organizing and managing the messages.
 - HTTP is a protocol that allows a user agent to access and manipulate the messages from a mail server using a web browser. HTTP uses TCP as the transport layer protocol and port 80 as the well-known port number. HTTP has a client-server model, where the user agent is the client and the mail server is the server. The user agent sends

File Transfer Protocol in application layer

- File Transfer Protocol (FTP) is an application layer protocol that is used to transfer the files between the local devices (PC, smartphone, etc.) to a server. It transfers both text and binary files over the Internet.
- FTP runs on the top of TCP, like HTTP. To transfer a file, 2 TCP connections are used by FTP in parallel: control connection and data connection.
- Control connection: It is used to send commands from the client to the server and receive replies from the server. It is established between the client port 21 and the server port 21. It remains open throughout the FTP session.
- Data connection: It is used to send the actual file data from the server to the client. It is established between the client port 20 and the server port 20. It is opened and closed for each file transfer.
- FTP uses two modes of operation: active mode and passive mode. In active mode, the server initiates the data connection to the client. In passive mode, the client initiates the data connection to the server.
- FTP supports three types of file transfer: ASCII mode, binary mode, and auto mode. ASCII mode is used to transfer text files, binary mode is used to transfer binary files, and auto mode is used to automatically detect the file type and choose the appropriate mode.
- FTP supports four types of file access: anonymous, basic, account, and secure. Anonymous access allows anyone to log in to the server without a username and password. Basic access requires a username and password to log in to the server. Account access requires a username, password, and an account name to log in to the server. Secure access requires a username, password, and an encryption method to log in to the server.
- FTP supports various commands to manipulate files and directories on the server, such as LIST, RETR, STOR, DELE, MKD, RMD, etc.
- FTP is widely used for uploading and downloading files from web servers, sharing files among users, and backing up data to remote servers.
- FTP has some limitations and drawbacks, such as lack of encryption, firewall issues, bandwidth consumption, and file size restrictions.

Mnemonics and learning tricks

- To remember the port numbers of FTP, you can use the following mnemonic: FTP uses **21** to control and **20** to send data.
- To remember the modes of FTP, you can use the following acronym: FTP can be **A**ctive or **P**assive.
- To remember the types of file transfer, you can use the following acronym: FTP can transfer files in **A**SCII, **B**inary, or **A**uto mode.
- To remember the types of file access, you can use the following acronym: FTP can access files **A**nonymously, **B**asically, with an **A**ccount, or **S**ecurely.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want

to learn or discuss. You have asked me to write on the topic of remote login in application layer. Here is my response:

Remote login in application layer

- Remote login is a service that allows a user to access a remote system or network over the internet using a terminal emulator or a command-line interface.
- Remote login is an example of an application layer protocol, which is the highest layer in the TCP/IP model or the seventh layer in the OSI model. Application layer protocols define the rules and formats for exchanging data between applications on different hosts.
- Some of the common remote login protocols are:
 - Telnet: A standard protocol for remote login that uses TCP port 23. It is an insecure protocol that sends data in plain text, making it vulnerable to eavesdropping and tampering.
 - Secure Shell (SSH): A secure protocol for remote login that uses encryption and authentication to protect data from unauthorized access. It uses TCP port 22 by default. It can also support file transfer, port forwarding, and tunneling.
 - Remote Desktop Protocol (RDP): A proprietary protocol for remote login that allows a user to access the graphical user interface (GUI) of a remote system. It uses TCP port 3389 by default. It can also support audio, video, printing, and clipboard functions.
- Some of the advantages of remote login are:
 - It enables remote administration and management of systems and networks, reducing the need for physical access and travel.
 - It allows users to access resources and applications that are not available on their local system, such as specialized software, databases, or servers.
 - It facilitates collaboration and communication among users who are geographically dispersed, such as teleworkers, students, or researchers.
- Some of the disadvantages of remote login are:
 - It poses security risks, such as unauthorized access, data theft, or malware infection, if the remote system or network is not properly secured or monitored.
 - It depends on the availability and reliability of the internet connection, which may affect the performance and quality of the remote login session.
 - It may cause compatibility issues, such as different operating systems, software versions, or screen resolutions, between the local and remote systems.
- Some of the applications of remote login are:
 - System administration: Remote login allows system administrators to perform tasks such as installing, updating, or troubleshooting soft-

ware, configuring settings, or monitoring performance on remote systems or networks.

- Education: Remote login allows students and teachers to access educational resources and applications, such as online courses, virtual labs, or simulations, that are hosted on remote servers or systems.
- Business: Remote login allows employees and clients to access business applications and data, such as email, documents, or databases, that are stored on remote servers or systems.

Network management in application layer

- Network management is the process of configuring, monitoring, and managing the performance of a network .
- The application layer is the topmost layer in the Open System Interconnection (OSI) model, which provides the interface between the network and the user applications .
- The application layer handles network services, such as file and printing, name resolution, and redirector services. Name resolution is the process of mapping an IP address to a human-readable name, such as a domain name.
- The application layer also supports various network management functions, such as:
 - Network monitoring: collecting and analyzing data on the network performance, such as traffic, errors, availability, and utilization.
 - Network configuration: setting up and modifying the network parameters, such as IP addresses, routing tables, security policies, and access control lists.
 - Network troubleshooting: detecting and resolving network problems, such as connectivity, latency, congestion, and security breaches.
 - Network optimization: improving the network efficiency, reliability, and quality of service, such as load balancing, bandwidth allocation, and traffic shaping.
- Some examples of network management protocols that operate at the application layer are:
 - Simple Network Management Protocol (SNMP): a widely used protocol for collecting and organizing information about network devices, such as routers, switches, servers, and printers.
 - Common Management Information Protocol (CMIP): a protocol for managing network elements that support the OSI management framework, such as telecommunications equipment and network services.
 - Remote Monitoring (RMON): a protocol for monitoring network traffic and statistics from remote devices, such as probes and agents.
 - Network Configuration Protocol (NETCONF): a protocol for installing, manipulating, and deleting the configuration of network devices, such as firewalls, load balancers, and VPNs.
- Some advantages of network management in the application layer are:

- It provides a high-level and user-friendly interface for network administration and operation.
- It allows interoperability and compatibility among different network devices and applications, regardless of their underlying hardware and software.
- It enables the integration and coordination of various network services and functions, such as security, quality of service, and mobility.
- Some disadvantages of network management in the application layer are:
 - It may introduce additional overhead and complexity to the network communication, such as processing, encoding, and decoding of messages.
 - It may be vulnerable to attacks and breaches, such as spoofing, tampering, and denial of service, if not properly secured and authenticated.
 - It may not be able to handle low-level and fine-grained network management tasks, such as packet filtering, switching, and routing.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have asked me to write on the topic of data compression in application layer. Here is my response:

Data compression in application layer

- Data compression is the process of reducing the size of data by using algorithms that eliminate redundancy or irrelevant information.
- Data compression can be performed at different layers of the network stack, such as the physical layer, the data link layer, the transport layer, or the application layer.
- Data compression at the application layer is the compression of data by the application itself, before sending or receiving it over the network.
- Data compression at the application layer has the following advantages:
 - It can reduce the bandwidth consumption and the transmission time of data over the network, which can improve the performance and efficiency of the application.
 - It can save the storage space and the memory usage of the data on the sender and the receiver side, which can reduce the cost and the complexity of the application.
 - It can allow the application to use more sophisticated and customized compression algorithms that are tailored to the specific type and format of the data, which can achieve higher compression ratios and better quality of the data.
- Data compression at the application layer has the following disadvantages:
 - It can increase the processing overhead and the latency of the data on

- the sender and the receiver side, which can degrade the performance and the responsiveness of the application.
 - It can introduce errors or distortions in the data due to the lossy nature of some compression algorithms, which can affect the accuracy and the reliability of the application.
 - It can require the compatibility and the interoperability of the compression algorithms and the formats used by the sender and the receiver, which can limit the flexibility and the scalability of the application.
- Data compression at the application layer can be classified into two types: lossless and lossy.
 - Lossless compression is the compression of data that preserves the original information and allows the exact reconstruction of the data after decompression. Lossless compression is suitable for data that requires high fidelity and integrity, such as text, code, or binary files.
 - Lossy compression is the compression of data that discards some information and allows the approximate reconstruction of the data after decompression. Lossy compression is suitable for data that can tolerate some degradation and distortion, such as audio, video, or images.
- Some examples of data compression algorithms and formats used at the application layer are:
 - ZIP, GZIP, BZIP2, and 7-ZIP are lossless compression algorithms and formats that can compress various types of data, such as text, code, or binary files. They use techniques such as Huffman coding, Lempel-Ziv coding, Burrows-Wheeler transform, and arithmetic coding to reduce the redundancy and the entropy of the data.
 - MP3, AAC, OGG, and FLAC are lossy or lossless compression algorithms and formats that can compress audio data, such as music or speech. They use techniques such as psychoacoustic modeling, quantization, transform coding, and subband coding to reduce the perceptual irrelevance and the redundancy of the audio data.
 - JPEG, PNG, GIF, and BMP are lossy or lossless compression algorithms and formats that can compress image data, such as photos or graphics. They use techniques such as color space conversion, run-length encoding, differential coding, and discrete cosine transform to reduce the spatial and the spectral redundancy of the image data.
 - MPEG, H.264, VP9, and AV1 are lossy compression algorithms and formats that can compress video data, such as movies or animations. They use techniques such as motion estimation, motion compensation, block-based coding, and entropy coding to reduce the temporal and the spatial redundancy of the video data.
- A mnemonic to remember the types of data compression at the application

layer is:

- **Lossless** compression preserves the **L**etter of the data.
- **Lossy** compression preserves the **L**ook of the data.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have asked me to write on the topic of cryptography in application layer. Here is my response:

Cryptography in application layer

- Cryptography in application layer refers to the use of cryptographic techniques and protocols to provide security and privacy for applications that run on top of the network layer, such as web browsers, email clients, instant messaging, etc.
- Cryptography in application layer can be divided into two categories: end-to-end encryption and link encryption.
 - End-to-end encryption means that the data is encrypted and decrypted only by the sender and the receiver of the communication, and not by any intermediate nodes or servers. This ensures that the data is protected from eavesdropping, tampering, or modification by third parties. Examples of end-to-end encryption protocols are PGP (Pretty Good Privacy) for email, Signal for instant messaging, and HTTPS (Hypertext Transfer Protocol Secure) for web browsing.
 - Link encryption means that the data is encrypted and decrypted at each hop of the communication, by the nodes or servers that relay the data. This ensures that the data is protected from unauthorized access or interception by the nodes or servers themselves, but not from the endpoints of the communication. Examples of link encryption protocols are SSL (Secure Sockets Layer) and TLS (Transport Layer Security) for web browsing, SMTPS (Simple Mail Transfer Protocol Secure) and IMAPS (Internet Message Access Protocol Secure) for email, and IPsec (Internet Protocol Security) for network layer security.
- Cryptography in application layer can provide various security services, such as confidentiality, integrity, authentication, non-repudiation, and access control.
 - Confidentiality means that the data is hidden from unauthorized parties, and can only be read by the intended recipients. Encryption algorithms, such as AES (Advanced Encryption Standard) and RSA (Rivest-Shamir-Adleman), are used to transform the data into an unreadable form, and decryption algorithms are used to reverse the process.
 - Integrity means that the data is protected from unauthorized modification, deletion, or insertion, and can be verified by the recipients. Hash functions, such as SHA (Secure Hash Algorithm) and MD5 (Message Digest 5), are used to generate a unique fingerprint of the

data, and digital signatures, such as DSA (Digital Signature Algorithm) and ECDSA (Elliptic Curve Digital Signature Algorithm), are used to sign the data with the sender's private key and verify it with the sender's public key.

- Authentication means that the identity of the sender and the receiver of the data can be confirmed and verified. Symmetric key algorithms, such as DES (Data Encryption Standard) and 3DES (Triple Data Encryption Standard), are used to generate and share a secret key between the sender and the receiver, and asymmetric key algorithms, such as RSA and ECC (Elliptic Curve Cryptography), are used to generate and exchange public and private keys between the sender and the receiver.
- Non-repudiation means that the sender and the receiver of the data cannot deny their involvement in the communication. Digital signatures, as mentioned above, are used to provide proof of origin and proof of delivery of the data, and timestamps, such as TSP (Time Stamp Protocol), are used to provide proof of existence and proof of order of the data.
- Access control means that the data can only be accessed by authorized parties, and can be restricted or revoked by the owner of the data. Encryption algorithms, as mentioned above, are used to prevent unauthorized access to the data, and access control mechanisms, such as ACL (Access Control List) and RBAC (Role-Based Access Control), are used to grant or deny access to the data based on the identity, role, or attributes of the parties.

Basic concepts of Cryptography in application layer

- Cryptography is the science of securing communications from adversaries by using mathematical techniques and algorithms.
- Cryptography can provide confidentiality, integrity, authenticity, and non-repudiation to the data transmitted or stored in applications.
- Cryptography can be classified into two main types: symmetric and asymmetric.
- Symmetric cryptography uses the same key for both encryption and decryption of data. The key must be shared securely between the sender and the receiver. Examples of symmetric algorithms are AES, DES, and RC4.
- Asymmetric cryptography, also known as public key cryptography, uses a pair of keys for encryption and decryption of data. One key is public and can be shared with anyone, while the other key is private and must be kept secret. The public key can be used to encrypt data, which can only be decrypted by the private key, or vice versa. Examples of asymmetric algorithms are RSA, ECC, and DSA.

- Cryptography can be applied in various layers of the network stack, such as the physical, data link, network, transport, and application layers. The application layer is the closest to the user and provides end-to-end security for the data.
- Some examples of cryptographic applications in the application layer are:
 - Secure Sockets Layer (SSL) and Transport Layer Security (TLS): These protocols provide secure communication between web browsers and web servers, using both symmetric and asymmetric cryptography. They also support digital certificates, which are used to verify the identity of the web servers and prevent man-in-the-middle attacks.
 - Secure Shell (SSH): This protocol provides secure remote access to servers and devices, using both symmetric and asymmetric cryptography. It also supports authentication, encryption, and integrity of the data.
 - Pretty Good Privacy (PGP) and GNU Privacy Guard (GPG): These are applications that provide secure email communication, using both symmetric and asymmetric cryptography. They also support digital signatures, which are used to verify the identity and integrity of the sender and the message.
 - Bitcoin and other cryptocurrencies: These are digital currencies that use cryptography to create and verify transactions, using both symmetric and asymmetric cryptography. They also use hash functions, which are mathematical functions that map any input to a fixed-length output, to create a public ledger of transactions, called the blockchain.